

SUMMER WORKSHOP IN MATHEMATICS

The Probabilistic Method

A five-day workshop for high-school students



Workshop Notes
(Lectures, Worked Examples, and Problems)

City of Lights
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“The probabilistic method has now developed to a point where it would be hopeless even to attempt to give a complete coverage of the area in one paper.” — Paul Erdős

About these notes. This document accompanies a five-day workshop on the probabilistic method aimed at mathematically curious 9th–10th standard students. Each chapter is one full day’s worth of material (about three hours of class time) and ends with a problem set that students take home, work on overnight, and return the next morning.

The problem sets are also available as two companion documents: `WorkshopProblems.pdf` (problems only, for day-by-day distribution to students) and `WorkshopSolutions.pdf` (the same problem sets, with worked solutions).

CONTENTS

1	Day 1 — Foundations of Probability	3
1.1	Sample Spaces and Events	4
1.2	Probability Measures	6
1.3	The Uniform Probability Measure	9
1.4	Independence of Events	11
1.5	Putting it Together	12
	Problem Set 1	13
2	Day 2 — Random Variables, Expectation, and Linearity	15
2.1	Random Variables	15
2.2	Expectation	16
2.3	Indicator Random Variables	19
2.4	Linearity of Expectation	20
2.5	A First Application: The Degree-Sum Formula	23
2.6	The Key Move: Computing Expectations	24
2.7	Independence of Random Variables (Optional)	26
	Problem Set 2	28
A	A Primer on Graphs	31
A.1	Graphs, Vertices, and Edges	31
A.2	Degree and the Handshake Lemma	32
A.3	Some Standard Graphs	33
A.4	Subgraphs and Complements	35
A.5	Cliques and Independent Sets	37
A.6	Edge Colourings	37
A.7	Glossary at a Glance	38

CHAPTER 1

DAY 1 — FOUNDATIONS OF PROBABILITY

The goal of this first day is to build, from scratch, the language we will need for the rest of the week: *sample spaces*, *events*, and *probability measures*. Although the workshop is about a powerful proof technique — the probabilistic method — the technique relies on a small handful of ideas that we now make precise.

A piece of advice before we begin. The single most common error in elementary probability is to start computing before deciding *what experiment* one is computing about. We will be obsessive about this. Every probability we compute will be tied to an explicit sample space and an explicit probability measure — this discipline is what allows us, later in the week, to use probability to prove results that have nothing to do with chance at all.

1.1 Why we need a formal language. Before we write down any definitions, let us see what goes wrong when we do not. Suppose two students are asked: *a fair coin is tossed twice; what is the chance that both tosses show heads?* The first student reasons, “There are three possible outcomes — two heads, two tails, or one of each — so the answer is $1/3$.” The second student insists, “No, there are four equally likely outcomes — HH, HT, TH, TT — so the answer is $1/4$.” Both arguments sound plausible if read quickly. Both cannot be correct. Yet neither student has done anything obviously foolish: each has listed what they take to be “the possible outcomes” and counted favourable ones. The disagreement is not about arithmetic. It is about *what experiment is being performed in the first place*, and which list of outcomes is the honest one. Once we agree to record the two tosses in order (so the four outcomes HH, HT, TH, TT form the right list, and each is equally likely), the answer is forced: $1/4$. The first student’s list was not wrong because the arithmetic was wrong; it was wrong because it grouped two genuinely different outcomes — the one where the first toss was a head and the second a tail, and the reverse — into a single entry, while still pretending each entry was equally likely.

This small disagreement is the whole story in miniature. The phrase “the probability of X ” is, on its own, only half a sentence; it is completed only when we also commit to *which experiment* X refers to and *how likelihoods are assigned* to the outcomes of that experiment. Change either ingredient and the same English words can yield a different number.

The trouble grows worse the moment the words “at random” enter the picture. Consider this beautiful old problem: pick a chord of a circle “at random”; what is the probability that the chord is longer than the side of the equilateral triangle inscribed in that circle? Three perfectly

reasonable interpretations of “random chord” give three different answers.

- Pick the two endpoints of the chord uniformly and independently on the circle. The probability turns out to be $1/3$.
- Pick a radius of the circle, and then pick a point on that radius uniformly; let the chord be the one perpendicular to the radius at that point. The probability is $1/2$.
- Pick the midpoint of the chord uniformly from the disc. The probability is $1/4$.

(Working out the three numbers is a pleasant exercise once we have the right vocabulary — we will not need them today.) None of these three procedures is wrong: they are simply *three different experiments*, all wearing the same English costume. Throwing a needle, spinning a spinner against a diameter, and tossing a dart at the disc are all natural physical realisations of “a random chord”, but they realise it differently. The question “what is the probability that a random chord is longer than the side of the inscribed triangle?” has no answer until the questioner says *which* random chord.

Even classroom-scale phrases hide the same trap. “Pick a student at random from a class of thirty” is ambiguous until we specify the procedure: drawing a roll number from a bowl uniformly, picking the first person to walk in tomorrow morning, or weighting students by some criterion we did not bother to write down. Two procedures that we casually describe with the same English sentence will, in general, place different probabilities on the same event.

The escape from this fog is not cleverness; it is discipline. Before we calculate anything, we will commit, in writing, to two ingredients: a set Ω listing all conceivable outcomes of the experiment, and a rule $\mathbb{P}$ that assigns to each subset of Ω a number in $[0, 1]$ representing the likelihood that the outcome lies in that subset. Once these are fixed, every probability becomes a well-defined number, and disagreements like the one between our two students reduce to disagreements about which Ω and which $\mathbb{P}$ correctly model the experiment at hand — a far more honest place to argue. The rest of Day 1 makes this discipline precise, beginning with the language of sample spaces and events.

1.1 SAMPLE SPACES AND EVENTS

1.2 The setup. With the discipline of the previous point in mind, we now collect the formal vocabulary we will use throughout the workshop: experiment, outcome, sample space, and event.

Definition. A **random experiment** is any procedure whose outcome is not predetermined. The set of all conceivable outcomes is called the **sample space** and is usually denoted Ω (capital Omega); its elements are called **outcomes**. Any subset $A \subseteq \Omega$ is called an **event**, and we say that the event A *occurs* when the actual outcome of the experiment lies in A .

We always insist that Ω is described *before* any computation is attempted. A handful of small examples will fix the vocabulary; in each, we describe the experiment in words, write down Ω explicitly, and name a few events as subsets of Ω .

- (i) *Roll a fair six-sided die once.* The sample space is

$$\Omega = \{1, 2, 3, 4, 5, 6\}.$$

The event “the roll is even” is the subset $A = \{2, 4, 6\} \subseteq \Omega$, and the event “the roll is at least 5” is $B = \{5, 6\}$. The event “the roll is a prime” is $\{2, 3, 5\}$.

- (ii) *Toss a coin three times in a row, recording the result of each toss in order.* Then

$$\Omega = \{\mathbf{H}, \mathbf{T}\}^3 = \{\mathbf{HHH}, \mathbf{HHT}, \mathbf{HTH}, \mathbf{HTT}, \mathbf{THH}, \mathbf{THT}, \mathbf{TTH}, \mathbf{TTT}\},$$

an 8-element set. The event “at least two heads” is $\{HHH, HHT, HTH, THH\}$; the event “the first toss is a tail” is $\{THH, THT, TTH, TTT\}$; the event “all three tosses are the same” is $\{HHH, TTT\}$.

- (iii) *Draw a single card from a standard 52-card deck.* The sample space Ω is the set of all 52 cards, so $|\Omega| = 52$. The event “the card is a heart” is the 13-card subset of hearts; “the card is a face card” is the 12-card subset consisting of the four jacks, four queens, and four kings; “the card is a red ace” is the 2-card subset $\{A\heartsuit, A\diamondsuit\}$.
- (iv) *Roll two distinguishable dice — say, one red and one blue — and record the ordered pair (a, b) of results.* Then

$$\Omega = \{1, \dots, 6\}^2,$$

a 36-element set. The event “the sum is 7” is

$$\{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$$

“Both dice show the same number” is

$$\{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)\}$$

“The red die is strictly larger” is the 15-element set $\{(a, b) : a > b\}$.

- (v) *A family has two children; record their genders in birth order.* Then

$$\Omega = \{B, G\}^2 = \{BB, BG, GB, GG\}$$

The event “both children are girls” is $\{GG\}$; “at least one child is a boy” is $\{BB, BG, GB\}$; “the two children have different genders” is $\{BG, GB\}$.

1.3 Combining events. Because events are subsets of Ω , we can combine them with the usual set operations. To make each definition concrete, fix the running example of a single roll of a fair die, with $\Omega = \{1, 2, 3, 4, 5, 6\}$, and the two events $A = \{2, 4, 6\}$ (“the roll is even”) and $B = \{5, 6\}$ (“the roll is at least 5”).

- *Union.* $A \cup B$ — “ A or B occurs” (or both): the outcome lies in A , in B , or in both. In the running example,

$$A \cup B = \{2, 4, 5, 6\},$$

the rolls that are even or at least 5 (with 6 qualifying on both counts).

- *Intersection.* $A \cap B$ — “ A and B both occur”: the outcome lies in both A and B simultaneously. In the running example,

$$A \cap B = \{6\},$$

since 6 is the only roll that is both even and at least 5.

- *Complement.* $A^c = \Omega \setminus A$ — “ A does not occur”: the outcome lies outside A . In the running example,

$$A^c = \{1, 3, 5\},$$

the odd rolls.

Two events A and B are **mutually exclusive** (or **disjoint**) when they cannot occur together, i.e. when $A \cap B = \emptyset$. The events A and B above are *not* mutually exclusive, since their intersection contains 6; for a pair that is, let $C = \{1, 3\}$ (“the roll is odd and less than 5”) and observe that $A \cap C = \emptyset$, because no roll is both even and odd.

Exercise 1.1.1. A fair coin is tossed twice, the results recorded in order. Write down the sample space Ω , and write the event “at least one toss is a head” as a subset of Ω .

Exercise 1.1.2. For a single roll of a fair die, let $A = \{2, 4, 6\}$ and $B = \{5, 6\}$. Write down each of $A \cup B$, $A \cap B$, and A^c , and decide whether A and B are mutually exclusive.

Exercise 1.1.3. A family has three children, whose genders are recorded in birth order. Write down the sample space Ω , and write the event “exactly two of the children are girls” as a subset of Ω .

1.2 PROBABILITY MEASURES

1.4 The probability measure. A probability measure on a finite sample space Ω is, at its heart, a rule that assigns a weight to each individual *outcome* of the experiment, recording how likely that outcome is. The probability of an *event* — a subset of Ω — is then nothing more than the total weight sitting inside it. We make this precise in two steps: first the function on outcomes, then the induced function on events.

Definition. Let Ω be a finite sample space. A **probability measure** on Ω is a function

$$\mathbb{P} : \Omega \rightarrow [0, 1]$$

whose values on the outcomes sum to 1:

$$\sum_{\omega \in \Omega} \mathbb{P}(\omega) = 1.$$

The number $\mathbb{P}(\omega) \in [0, 1]$ is called the **probability mass** (or simply the **probability**) of the outcome ω .

With masses on outcomes in hand, the probability of any event $A \subseteq \Omega$ is, by definition, the sum of the masses of its outcomes:

$$\mathbb{P}(A) := \sum_{\omega \in A} \mathbb{P}(\omega).$$

We follow the standard abuse of notation and write $\mathbb{P}$ both for the function on outcomes (with input $\omega \in \Omega$) and for the induced function on events (with input $A \subseteq \Omega$); context always makes clear which is meant. Note that the original mass on a single outcome ω is recovered as the probability of the singleton event $\{\omega\}$. Three consequences of the definition come up so often that we name them explicitly.

(P1) *The total probability is 1.* Directly from the defining sum,

$$\mathbb{P}(\Omega) = \sum_{\omega \in \Omega} \mathbb{P}(\omega) = 1.$$

In words: something always happens.

(P2) *Probabilities are non-negative.* For every event $A \subseteq \Omega$,

$$\mathbb{P}(A) = \sum_{\omega \in A} \mathbb{P}(\omega) \geq 0,$$

because each term in the sum is non-negative.

(P3) *Disjoint events add.* If A and B are disjoint events ($A \cap B = \emptyset$), then every outcome in $A \cup B$ belongs to exactly one of A or B , so splitting the sum gives

$$\mathbb{P}(A \cup B) = \sum_{\omega \in A \cup B} \mathbb{P}(\omega) = \sum_{\omega \in A} \mathbb{P}(\omega) + \sum_{\omega \in B} \mathbb{P}(\omega) = \mathbb{P}(A) + \mathbb{P}(B).$$

The same argument, applied repeatedly, gives the general form: for pairwise disjoint events $A_1, A_2, \dots, A_n$,

$$\mathbb{P}(A_1 \cup A_2 \cup \dots \cup A_n) = \mathbb{P}(A_1) + \mathbb{P}(A_2) + \dots + \mathbb{P}(A_n).$$

Almost every calculation in the workshop rests on (P1) and (P3); (P2) is so transparent from the definition that we record it mostly for completeness. We will refer back to these three properties by their labels throughout.

1.5 What does the probability of an event mean? The definition we have just given tells us how to *manipulate* probabilities — they are non-negative numbers, they sum to 1, they add over disjoint events — but it is silent on what the number $\mathbb{P}(A)$ is supposed to *represent* in the world. A reader who has never seen a probability measure before is well within their rights to ask: in what sense is the number $1/6$ the “right” probability of rolling a 3 on a fair die? Why this number and not some other?

The most popular answer is the **frequentist interpretation**.¹ Imagine repeating the same experiment under identical conditions a large number of times — N tosses of a coin, N rolls of a die, N random draws of a card — and recording, for each event A , the fraction

$$\hat{p}_N(A) = \frac{\#\{\text{trials in which } A \text{ occurred}\}}{N}.$$

The frequentist interpretation says: *the number $\mathbb{P}(A)$ is the long-run value of $\hat{p}_N(A)$ as N grows large.* In other words, $\mathbb{P}(A)$ is what the empirical fraction $\hat{p}_N(A)$ converges to as we run the experiment more and more times. Toss a fair coin 1000 times and you will see roughly 500 heads; toss it a million times and you will see roughly 500 000. The fraction of heads stabilises near $1/2$, and we declare $\mathbb{P}(\text{heads}) = 1/2$.

Reading the three properties (P1), (P2), (P3) through this lens, each of them becomes inevitable rather than imposed.

- (i) *Total probability is 1.* Every trial produces exactly one outcome, so the trial counts of any partition of Ω into singletons sum to N . Dividing by N , the empirical fractions across Ω sum to 1; passing to the long-run limit gives $\mathbb{P}(\Omega) = 1$, which is (P1).
- (ii) *Probabilities are non-negative.* An empirical fraction $\hat{p}_N(A)$ is a count divided by N , hence non-negative, and so its limit is non-negative too. This is (P2).
- (iii) *Disjoint events add.* If A and B are disjoint, then no single trial is counted in both: the trials in which $A \cup B$ occurred are precisely the trials in A plus those in B . Dividing by N , $\hat{p}_N(A \cup B) = \hat{p}_N(A) + \hat{p}_N(B)$ exactly. Passing to the limit gives (P3).

So the axioms are not arbitrary; they are exactly the rules that empirical fractions automatically satisfy, lifted to the long-run limit.

A word of caution. The interpretation describes *what* $\mathbb{P}(A)$ *means*; it is not a definition we can use to *compute* $\mathbb{P}(A)$, because we never actually run an experiment infinitely many times. In practice, one models the experiment by writing down a sample space and a probability measure

¹It is not the only answer. A second tradition, the *subjective* or *Bayesian* school, reads $\mathbb{P}(A)$ as a numerical measure of one’s personal degree of belief that A will occur. The two interpretations rarely lead to different calculations, and for the workshop we will be happy with the frequentist picture.

that one believes describes it correctly (often by symmetry, as for the fair coin or the fair die), and then computes with that model. The frequentist picture is the bridge between the model and reality — it explains why a number on paper deserves the name “probability”. The rest of the workshop will live almost entirely on the modelling side of the bridge; we will write down $\mathbb{P}$, manipulate it, and prove theorems about it, leaning on the frequentist intuition only when we need to remember what the symbols are about.

One final note. Even within the frequentist picture, the claim that $\hat{p}_N(A) \rightarrow \mathbb{P}(A)$ as $N \rightarrow \infty$ is itself a theorem, not a definition — this is the content of the *law of large numbers*, which we will not prove in the workshop. We will, however, see a quantitative cousin of it on Day 5: Chebyshev’s inequality, which controls how close $\hat{p}_N(A)$ is to $\mathbb{P}(A)$ for any specific finite N .

1.6 Examples of probability measures. The definition becomes much clearer in action. In each example below we exhibit a sample space Ω and the mass $\mathbb{P}(\omega)$ assigned to each outcome, check that the masses sum to 1, and — where it adds something — use the identity $\mathbb{P}(A) = \sum_{\omega \in A} \mathbb{P}(\omega)$ to compute the probability of a representative event.

- (i) *The fair coin.* Take $\Omega = \{\mathbf{H}, \mathbf{T}\}$ with $\mathbb{P}(\mathbf{H}) = \mathbb{P}(\mathbf{T}) = 1/2$. The masses sum to $1/2 + 1/2 = 1$, so this is a valid probability measure. The event “the coin shows heads” is the singleton $\{\mathbf{H}\}$, with $\mathbb{P}(\{\mathbf{H}\}) = 1/2$.
- (ii) *A biased coin.* On the same sample space $\Omega = \{\mathbf{H}, \mathbf{T}\}$ we could equally well take $\mathbb{P}(\mathbf{H}) = 2/3$ and $\mathbb{P}(\mathbf{T}) = 1/3$; the masses still sum to 1, so this too is a probability measure on Ω . The lesson to take away is that $\mathbb{P}$ is its own ingredient, separate from Ω : a single sample space carries infinitely many distinct probability measures, one for each physical experiment we might have in mind.
- (iii) *A fair die.* On $\Omega = \{1, 2, 3, 4, 5, 6\}$, set $\mathbb{P}(k) = 1/6$ for every $k \in \Omega$. The masses sum to $6 \cdot (1/6) = 1$. For the event $A = \{2, 4, 6\}$ (“the roll is even”),

$$\mathbb{P}(A) = \mathbb{P}(2) + \mathbb{P}(4) + \mathbb{P}(6) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2}.$$

- (iv) *A loaded die.* Take the same sample space $\Omega = \{1, 2, 3, 4, 5, 6\}$, but now suppose the die has been weighted so that the face 6 comes up with probability $1/2$, and each of the other five faces with probability $1/10$. Check first that the masses sum correctly:

$$\mathbb{P}(1) + \mathbb{P}(2) + \mathbb{P}(3) + \mathbb{P}(4) + \mathbb{P}(5) + \mathbb{P}(6) = 5 \cdot \frac{1}{10} + \frac{1}{2} = \frac{1}{2} + \frac{1}{2} = 1.$$

The same event $A = \{2, 4, 6\}$ (“the roll is even”) now has probability

$$\mathbb{P}(A) = \mathbb{P}(2) + \mathbb{P}(4) + \mathbb{P}(6) = \frac{1}{10} + \frac{1}{10} + \frac{1}{2} = \frac{7}{10}.$$

This is exactly the same event, on exactly the same sample space, as in the previous example — but the probability has shifted from $1/2$ to $7/10$, because the underlying mass function changed. “The probability that the die shows an even number” is not a property of the die alone; it is a property of the pair (die, weighting).

- (v) *A degenerate measure.* On $\Omega = \{a, b, c\}$, the assignment $\mathbb{P}(a) = 1$, $\mathbb{P}(b) = 0$, $\mathbb{P}(c) = 0$ has masses summing to 1 and so is a perfectly legal probability measure — though a boring one, since it describes an experiment whose outcome is a with certainty. The event $\{b, c\}$ has probability $\mathbb{P}(b) + \mathbb{P}(c) = 0$: even though b and c appear in the sample space, this particular measure rules them out, and outcomes of mass zero contribute nothing to the probability of any event that contains them.

1.7 Consequences of the rules. A few simple consequences will be used constantly.

Elementary properties. Let $A, B \subseteq \Omega$ be events. Then

- (i) $\mathbb{P}(\emptyset) = 0$;
- (ii) $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$;
- (iii) if $A \subseteq B$, then $\mathbb{P}(A) \leq \mathbb{P}(B)$;
- (iv) $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$;
- (v) $\mathbb{P}(A \cup B) \leq \mathbb{P}(A) + \mathbb{P}(B)$ (**union bound**).

Proof. (i) The empty set is disjoint from itself, so by (P3), $\mathbb{P}(\emptyset) = \mathbb{P}(\emptyset \cup \emptyset) = 2\mathbb{P}(\emptyset)$, forcing $\mathbb{P}(\emptyset) = 0$. (ii) Since A and A^c are disjoint and $A \cup A^c = \Omega$, we have $\mathbb{P}(A) + \mathbb{P}(A^c) = 1$. (iii) Write $B = A \cup (B \setminus A)$ as a disjoint union; then $\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A) \geq \mathbb{P}(A)$. (iv) Decompose $A \cup B$ as the disjoint union $A \cup (B \setminus A)$, and decompose B as the disjoint union $(B \setminus A) \cup (A \cap B)$. Adding gives the claim. (v) Follows from (iv) since $\mathbb{P}(A \cap B) \geq 0$. ■

The union bound (v) extends to any number of events: $\mathbb{P}(A_1 \cup \dots \cup A_n) \leq \mathbb{P}(A_1) + \dots + \mathbb{P}(A_n)$. Despite (or because of) its simplicity, the union bound will be one of the workhorses of the workshop.

Exercise 1.2.1. A die is loaded so that the face 6 comes up with probability $1/2$, and each of the other five faces with probability $1/10$. Find the probability that the roll is odd, and the probability that the roll is at least 5.

Exercise 1.2.2. Let A and B be events with $\mathbb{P}(A) = 0.6$, $\mathbb{P}(B) = 0.5$, and $\mathbb{P}(A \cap B) = 0.2$. Find $\mathbb{P}(A \cup B)$ and $\mathbb{P}(A^c)$.

Exercise 1.2.3. Using the two-event union bound $\mathbb{P}(A \cup B) \leq \mathbb{P}(A) + \mathbb{P}(B)$, show that for any three events A , B , and C ,

$$\mathbb{P}(A \cup B \cup C) \leq \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C).$$

1.3 THE UNIFORM PROBABILITY MEASURE

1.8 Equally likely outcomes. When the sample space Ω is finite and we model each outcome as “equally likely”, the probability measure is uniquely determined: each singleton $\{\omega\}$ must receive probability $1/|\Omega|$. From (P3), the probability of any event is then

$$\mathbb{P}(A) = \frac{|A|}{|\Omega|} = \frac{\#\{\text{favourable outcomes}\}}{\#\{\text{total outcomes}\}}.$$

This is the **uniform probability measure**, sometimes also called the *equiprobable model*. Almost every problem on the first day will use this model.

Example 1.9. Roll a fair six-sided die twice; record the ordered pair (a, b) of results. Then $\Omega = \{1, \dots, 6\}^2$, so $|\Omega| = 36$. The event $A = \{a + b = 7\}$ contains the six outcomes $(1, 6), (2, 5), \dots, (6, 1)$, so under the uniform measure $\mathbb{P}(A) = 6/36 = 1/6$.

Example 1.10. A box contains 5 red and 3 blue balls; two balls are drawn without replacement. What is the probability that both are red? Take Ω to be the set of unordered pairs of balls; $|\Omega| = \binom{8}{2} = 28$. The number of unordered pairs of red balls is $\binom{5}{2} = 10$. So the probability is $10/28 = 5/14$.

There is a second, equally valid sample space: ordered pairs of distinct balls. Then $|\Omega| = 8 \cdot 7 = 56$ and the favourable count is $5 \cdot 4 = 20$, giving the same answer $20/56 = 5/14$. This is reassuring: the answer should not depend on whether we record order, provided we are consistent.

Example 1.11 (A three-card deck, by hand). Before trusting the factorials, it helps to see “every ordering equally likely” on a deck small enough to write out in full. Take three distinct cards — an Ace, a King, and a Queen — and shuffle them uniformly. The sample space is the set of all $3! = 6$ orderings,

$$\text{AKQ, AQK, KAQ, KQA, QAK, QKA,}$$

each with probability $1/6$ under the uniform measure.

What is the probability that the Ace is on top? Two of the six orderings, AKQ and AQK, begin with the Ace, so

$$\mathbb{P}(\text{Ace on top}) = \frac{2}{6} = \frac{1}{3}.$$

This is exactly the $1/n$ we will find for a uniform shuffle of n cards (here $n = 3$); the $1/52$ of the next example is the same statement on a larger deck.

What is the probability that the Ace comes before the King? The favourable orderings are AKQ, AQK, and QAK — three of the six — so the probability is $3/6 = 1/2$. And it must be: nothing in a fair shuffle prefers “Ace before King” to “King before Ace”, so the two are equally likely. We will lean on this symmetry again when asking which of two cards a shuffle places first.

Example 1.12 (A shuffled deck). Shuffle a standard 52-card deck so that every ordering is equally likely. The sample space is $\Omega = S_{52}$, the set of all orderings of the deck, with $|\Omega| = 52!$ under the uniform measure. A uniform shuffle is the canonical equiprobable model on a large Ω , and most questions about it dissolve into a permutation count.

What is the probability that the top card is the Ace of Spades? An ordering puts $A\spadesuit$ on top exactly when the remaining 51 cards fill the other 51 positions in some order, and there are $51!$ such orderings. Hence

$$\mathbb{P}(\text{top card is } A\spadesuit) = \frac{51!}{52!} = \frac{1}{52}.$$

There is nothing special about the spade ace, or about the top: by the same count, *any* named card is equally likely to occupy *any* named position, each with probability $1/52$. This is the symmetry of a fair shuffle, made precise by counting rather than asserted.

What is the probability that the top two cards are both aces? Count the orderings whose first two positions hold aces. There are $4 \cdot 3$ ways to place an ordered pair of aces in the top two slots, and $50!$ ways to arrange the remaining cards below them, for $4 \cdot 3 \cdot 50!$ favourable orderings. Therefore

$$\mathbb{P}(\text{top two cards are aces}) = \frac{4 \cdot 3 \cdot 50!}{52!} = \frac{12}{52 \cdot 51} = \frac{1}{221}.$$

1.13 A Worked Example: The Birthday Problem. Probability under the uniform model reduces to counting, and we will lean on a small set of standard counting principles — multiplication, permutations, k -permutations, and combinations (binomial coefficients) — throughout this workshop. These are collected, with their derivations, in Appendix ??; readers who are not yet comfortable with any of the four should pause and skim that appendix before continuing. A first application of these principles, big enough to be interesting and small enough to be solved in one move, is the famous birthday problem.

A classroom contains 30 students. Assuming each student’s birthday is equally likely to be any of 365 days, and birthdays are chosen independently, what is the probability that some two students share a birthday? The cleanest sample space is $\Omega = \{1, \dots, 365\}^{30}$, with the uniform measure; an outcome is an ordered list of 30 birthdays, one per student. So $|\Omega| = 365^{30}$. The

complement of “some two share a birthday” is “all 30 birthdays are distinct”. Counting the ordered 30-tuples of *distinct* birthdays gives

$$365 \cdot 364 \cdot 363 \cdots 336 = \frac{365!}{335!}.$$

So

$$\mathbb{P}(\text{all distinct}) = \frac{365 \cdot 364 \cdots 336}{365^{30}} \approx 0.294,$$

and the probability of a shared birthday is about 0.706. The lesson, which will reappear: with 30 outcomes among 365 slots, “collisions” are far more likely than the size of the room alone might suggest.

Exercise 1.3.1. A fair die is rolled twice. Find the probability that the sum of the two rolls equals 5, and the probability that the sum is even.

Exercise 1.3.2. A box contains 4 red balls and 2 green balls. Two balls are drawn without replacement. Find the probability that both are red.

Exercise 1.3.3. A number is chosen uniformly at random from $\{1, 2, \dots, 10\}$. Find the probability that it is prime.

Exercise 1.3.4. A standard 52-card deck is shuffled uniformly at random. Find the probability that the bottom card is a spade, and the probability that the top two cards are both spades.

1.4 INDEPENDENCE OF EVENTS

1.14 Independence. Two events A and B are called **independent** if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B).$$

Intuitively: knowing whether A occurred does not change the probability that B occurs. The definition is what it is — a *multiplication rule for the joint probability* — and we will use it in that form, without philosophy.

More generally, events $A_1, A_2, \dots, A_n$ are **mutually independent** if for every subset of indices $i_1 < \dots < i_k$,

$$\mathbb{P}(A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k}) = \mathbb{P}(A_{i_1}) \mathbb{P}(A_{i_2}) \cdots \mathbb{P}(A_{i_k}).$$

(One must check the multiplication rule for *every* subset, not only for pairs.)

Example 1.15 (Position-fixed events in a random permutation). Let π be a uniformly random permutation of $\{1, 2, \dots, n\}$, with $n \geq 2$. For each i , let

$$A_i = \{\pi(i) = i\}$$

be the event “ i is a fixed point of π .” These events look as if they ought to be independent — the position $\pi(1)$ takes seems to have nothing to do with the position $\pi(2)$ takes — and yet they are not.

Work in $\Omega = S_n$ (all $n!$ permutations, with the uniform measure). The number of permutations fixing $\pi(1) = 1$ is $(n-1)!$ (we are free to permute the remaining $n-1$ values), so

$$\mathbb{P}(A_1) = \frac{(n-1)!}{n!} = \frac{1}{n}.$$

Similarly, the number of permutations fixing both $\pi(1) = 1$ and $\pi(2) = 2$ is $(n-2)!$, and so

$$\mathbb{P}(A_1 \cap A_2) = \frac{(n-2)!}{n!} = \frac{1}{n(n-1)}.$$

If A_1 and A_2 were independent, the joint probability would factor as $\mathbb{P}(A_1)\mathbb{P}(A_2) = 1/n^2$. Instead,

$$\frac{1}{n(n-1)} > \frac{1}{n^2}.$$

Knowing $\pi(1) = 1$ makes $\pi(2) = 2$ *more* likely, because once the value 1 has been used at position 1, the value 2 has one fewer competitor for position 2. The bias is small for large n but it is real, and it is exactly the kind of dependence that informal reasoning tends to miss.

1.16 A first useful identity. A typical use of mutual independence is when we want to argue that “none of a long list of bad things happens”. Applying the multiplication rule to the complementary events — i.e., to the events “each individual bad thing does *not* happen” — turns this into a product, which is often vastly easier to estimate than the original probability.

Probability of avoiding many events. If $A_1, \dots, A_n$ are mutually independent, then

$$\mathbb{P}\left(\bigcap_{i=1}^n A_i^c\right) = \prod_{i=1}^n \mathbb{P}(A_i^c) = \prod_{i=1}^n (1 - \mathbb{P}(A_i)).$$

Proof. It is a (standard but non-trivial) fact that if $A_1, \dots, A_n$ are mutually independent then so are $A_1^c, \dots, A_n^c$. The identity follows by applying the multiplication rule. ■

A useful consequence we will return to repeatedly: for any reals x_i ,

$$\prod_{i=1}^n (1 - x_i) \leq \exp\left(-\sum_{i=1}^n x_i\right),$$

since $1 - x \leq e^{-x}$ for all real x . So even when the union bound is too crude, an independence argument can produce sharp exponential decay.

Exercise 1.4.1. Events A and B are independent, with $\mathbb{P}(A) = 1/2$ and $\mathbb{P}(B) = 1/3$. Find $\mathbb{P}(A \cap B)$ and $\mathbb{P}(A \cup B)$.

Exercise 1.4.2. A fair coin is tossed three times, so that $\Omega = \{\text{H}, \text{T}\}^3$ carries the uniform measure. Let A be the event “the first toss is H” and B the event “the second toss is T”. By counting outcomes, determine whether A and B are independent.

Exercise 1.4.3. Let $A_1, \dots, A_n$ be mutually independent events, each of probability p . Write the probability that none of them occurs as a product, and use the inequality $1 - x \leq e^{-x}$ to give an upper bound for it.

1.5 PUTTING IT TOGETHER

1.17 The shape of every probability problem. Every problem we solve in this workshop — whether elementary or advanced — has the same shape:

- (1) **Choose a sample space** Ω that captures the experiment.
- (2) **Choose a probability measure** $\mathbb{P}$ on Ω (most often, the uniform measure on a cleverly chosen Ω).
- (3) **Identify the event** A whose probability you want.
- (4) **Compute** $\mathbb{P}(A)$, often by counting or by combining simpler probabilities using the rules above.

The *art* lies in steps (1) and (2): the right Ω can turn an unwieldy question into a one-line calculation. Step (4) is then mostly a careful count.

PROBLEM SET 1

The following problems are to be attempted before tomorrow's session. They consolidate today's definitions (sample spaces, events, the uniform measure, counting, and independence) and prepare the ground for the language of expectation, which we will introduce on Day 2.

Problem 1.5.1. A fair coin is tossed four times in a row.

- (a) Write down a sample space Ω for this experiment, and state the probability measure $\mathbb{P}$ on Ω .
- (b) Compute $\mathbb{P}(\text{exactly two heads})$.
- (c) Compute $\mathbb{P}(\text{the first toss is a head})$.
- (d) Compute $\mathbb{P}(\text{the first and last tosses agree})$.

Problem 1.5.2. Five distinct cards are drawn from a standard 52-card deck.

- (a) How many distinct 5-card hands are there?
- (b) What is the probability that all five cards are spades?
- (c) What is the probability that the hand contains exactly one ace?

Problem 1.5.3. Show, directly from the three rules (P1)–(P3), that for any events $A_1, A_2, \dots, A_n$,

$$\mathbb{P}\left(\bigcup_{i=1}^n A_i\right) \leq \sum_{i=1}^n \mathbb{P}(A_i).$$

This is the *union bound*; we will use it many times.

Problem 1.5.4. Twenty people are seated at random around a circular table. What is the probability that two particular people, Alice and Bob, sit next to each other?

Problem 1.5.5. Six six-sided dice are rolled simultaneously. Compute the probability that

- (a) all six show the same value;
- (b) all six values are different;
- (c) at least two dice show the same value.

Problem 1.5.6. Let $\Omega = \{1, 2, 3, 4\}$ with the uniform probability measure. Define the events

$$A = \{1, 2\}, \quad B = \{1, 3\}, \quad C = \{1, 4\}.$$

- (a) Show that A, B, C are *pairwise* independent.
- (b) Show that A, B, C are *not* mutually independent.

(This shows that pairwise independence is strictly weaker than mutual independence.)

Problem 1.5.7. A coin with probability p of heads is tossed n times independently. Express the probability that the resulting string contains at least one head. Now use the inequality $1 - x \leq e^{-x}$ to show that for any $p \in (0, 1)$, this probability exceeds $1 - e^{-pn}$.

This is our first “meet a threshold” calculation: for a small p , we need only $n \approx 1/p$ tosses before a head becomes very likely.

Problem 1.5.8. A standard 52-card deck is shuffled uniformly at random. What is the probability that the Ace of Spades appears earlier in the shuffled order than the Ace of Hearts?

Problem 1.5.9. Three friends — Alice, Bob, and Carol — leave their wallets at the door of a party. As they leave, an inattentive attendant returns the three wallets to the three friends in a uniformly random order.

- (a) Compute the probability that every friend gets their own wallet.
- (b) Compute the probability that *nobody* gets their own wallet.
- (c) Compute the probability that exactly one friend gets their own wallet.
- (d) Could exactly two friends get their own wallets? Explain.

Problem 1.5.10. The $2n$ players of a tennis tournament are paired up uniformly at random into n matches (so the sample space is the set of all ways to partition $\{1, 2, \dots, 2n\}$ into n unordered pairs, with the uniform measure on it). What is the probability that players 1 and 2 are paired against each other?

Problem 1.5.11. A fair coin is tossed five times, producing a string in $\{H, T\}^5$.

- (a) Compute the probability that the string contains no two consecutive Hs (i.e., no HH substring).
- (b) Let a_n denote the number of length- n strings in $\{H, T\}^n$ with no HH substring. By splitting on the last character, show that

$$a_n = a_{n-1} + a_{n-2} \quad \text{for } n \geq 3.$$

(Do you recognise the sequence?)

CHAPTER 2

DAY 2 — RANDOM VARIABLES, EXPECTATION, AND LINEARITY

Yesterday we developed the language of sample spaces and events. Today we add the workhorse on top of that language: the *random variable*, and its single most important quantity, the *expectation*. By the end of the day we will know the rule that drives almost every proof in the rest of the workshop — *linearity of expectation* — and will see it in action through one classical identity (the degree-sum formula).

2.1 RANDOM VARIABLES

2.1 Random variables. Sample spaces and probability measures, on their own, only let us talk about events — yes/no questions about an experiment. Most of the questions that interest us, however, ask for a *readout*: *how many heads*, *what sum*, *how large the largest piece*, *which vertex*, *what permutation*. The *random variable* packages such a readout into a single mathematical object.

Definition. Let $(\Omega, \mathbb{P})$ be a finite probability space, and let T be a set. A **random variable** on $(\Omega, \mathbb{P})$ with **target set** T is a function

$$X : \Omega \rightarrow T.$$

Intuitively, $X(\omega)$ is the readout the experiment produces when the outcome turns out to be ω . When $T = \mathbb{R}$ we call X a **real-valued** random variable. This is by far the most common case — and, as we will see in the next section, the only case in which the central notion of *expectation* is defined — so when we say “random variable” without further qualification, we mean real-valued.

A random variable comes with a host of derived events. Given any value $t \in T$ (or any subset $S \subseteq T$), the set $\{\omega \in \Omega : X(\omega) = t\}$ is itself an event in Ω — the event that, in this run of the experiment, X takes the value t . The shorthand

$$\mathbb{P}(X = t) \quad \text{stands for} \quad \mathbb{P}(\{\omega \in \Omega : X(\omega) = t\})$$

and similarly

$$\mathbb{P}(X \in S) \text{ stands for } \mathbb{P}(\{\omega : X(\omega) \in S\})$$

When X is real-valued, the further shorthands $\mathbb{P}(X \leq k)$, $\mathbb{P}(X < k)$, and so on are defined analogously, since the values of X are then numbers we can order. A random variable thus packages up a whole family of events — one for each question one might ask about X . A few examples will fix the notion.

- (i) *Number of heads in three fair coin tosses.* Take $\Omega = \{\mathbf{H}, \mathbf{T}\}^3$ with the uniform measure, and let $X(\omega)$ be the number of Hs in the string ω . Then X takes values in $\{0, 1, 2, 3\}$, with

$$\mathbb{P}(X = 0) = \frac{1}{8}, \quad \mathbb{P}(X = 1) = \frac{3}{8}, \quad \mathbb{P}(X = 2) = \frac{3}{8}, \quad \mathbb{P}(X = 3) = \frac{1}{8}$$

(one outcome maps to 0, three to 1, three to 2, one to 3).

- (ii) *Sum of two dice.* Take $\Omega = \{1, \dots, 6\}^2$ with the uniform measure, and let $S(\omega) = a + b$ where $\omega = (a, b)$. Then S takes values in $\{2, 3, \dots, 12\}$. Six of the 36 outcomes have $a + b = 7$, so $\mathbb{P}(S = 7) = 6/36 = 1/6$.
- (iii) *Maximum of three dice.* Take $\Omega = \{1, \dots, 6\}^3$ with the uniform measure, and let $M(\omega) = \max(a, b, c)$ where $\omega = (a, b, c)$. The event “ $M \leq k$ ” is precisely the set of triples in which every component is at most k , so $\mathbb{P}(M \leq k) = (k/6)^3$ for each $k \in \{1, \dots, 6\}$.
- (iv) *A constant random variable.* The function $C(\omega) = 17$ for every ω is also (somewhat dully) a random variable. Constants are random variables that happen to read the same number on every outcome; here $\mathbb{P}(C = 17) = 1$ and $\mathbb{P}(C = k) = 0$ for any $k \neq 17$.
- (v) *The outcome itself.* If $\Omega = \{1, \dots, 6\}$ is a single die roll, then the function $X(\omega) = \omega$ (“the number rolled”) is a random variable on Ω . Whenever the outcomes are themselves numbers, the identity function is the most natural random variable on the space.
- (vi) *A random vertex (not real-valued).* Let $G = (V, E)$ be a finite graph. Take $\Omega = V$ with the uniform measure $\mathbb{P}(\{v\}) = 1/|V|$, and let $U(\omega) = \omega$. Then $U : \Omega \rightarrow V$ is a random variable whose target set is V itself — a set of vertices, not a set of numbers. The event $\{U = v\}$ has probability $1/|V|$ for each $v \in V$, and we read this as “ U is a uniformly random vertex of G .” We will use exactly this construction in the proof of the degree-sum formula later in the chapter. Non-numerical readouts are useful: “a uniformly random permutation of $\{1, \dots, n\}$ ” and “a uniformly random vertex of a graph” are random variables in this broader sense, even though we cannot meaningfully ask for their averages.

Exercise 2.1.1. A fair coin is tossed twice, and X records the number of heads. List the values X can take, and for each value k write down $\mathbb{P}(X = k)$.

Exercise 2.1.2. Two fair dice are rolled, recorded as an ordered pair, so $\Omega = \{1, \dots, 6\}^2$. Let M be the larger of the two values shown. Find $\mathbb{P}(M \leq 4)$ and $\mathbb{P}(M = 6)$.

Exercise 2.1.3. With $\Omega = \{1, \dots, 6\}^2$ as above, let S be the sum of the two dice. Write the event $\{S \geq 10\}$ as an explicit subset of Ω , and find $\mathbb{P}(S \geq 10)$.

2.2 EXPECTATION

2.2 The expected value. The expectation of a random variable is the long-run *average* value it produces, when the experiment is repeated many times. On a finite probability space this average is exactly the weighted average of the values, with weights given by the probability masses.

Definition. Let X be a *real-valued* random variable on a finite probability space $(\Omega, \mathbb{P})$. The **expectation** (or **expected value**, or **mean**) of X is

$$\mathbb{E}[X] = \sum_{\omega \in \Omega} X(\omega) \cdot \mathbb{P}(\omega).$$

(Expectation requires that the values $X(\omega)$ be numbers we can add up and weight by probabilities, so we restrict to the real-valued case. For random variables with a non-numerical target set, the notion of average is undefined.)

Two remarks before any computation. First, the formal definition sums over the *outcomes* $\omega \in \Omega$, each contributing $X(\omega)$ weighted by its mass $\mathbb{P}(\omega)$. An equivalent “value-by-value” formulation, often more convenient in practice, is

$$\mathbb{E}[X] = \sum_x x \cdot \mathbb{P}(X = x),$$

where x ranges over the values that X actually takes. The two formulas agree because the second simply groups together those outcomes ω for which $X(\omega) = x$ and replaces them by a single term. Second, the expectation need not equal any value X actually takes; “average” and “typical” are different things. The very first example illustrates this.

- (i) *Fair die.* Let X be the number rolled on a fair six-sided die, so $\Omega = \{1, \dots, 6\}$ with each outcome of mass $1/6$. Then

$$\mathbb{E}[X] = \sum_{k=1}^6 k \cdot \frac{1}{6} = \frac{1+2+3+4+5+6}{6} = \frac{21}{6} = 3.5.$$

The die never shows the value 3.5 — the expectation is an average, not a possible outcome.

- (ii) *Three fair coins.* Let X be the number of heads in three fair coin tosses. From the previous section, X takes the values $\{0, 1, 2, 3\}$ with probabilities $\{1/8, 3/8, 3/8, 1/8\}$, so

$$\mathbb{E}[X] = 0 \cdot \frac{1}{8} + 1 \cdot \frac{3}{8} + 2 \cdot \frac{3}{8} + 3 \cdot \frac{1}{8} = \frac{12}{8} = 1.5.$$

Once again, this is a value X never takes.

- (iii) *A loaded die.* Take the loaded die from Day 1: $\Omega = \{1, \dots, 6\}$ with $\mathbb{P}(6) = 1/2$ and $\mathbb{P}(k) = 1/10$ for $k \in \{1, \dots, 5\}$. Let X be the value rolled. Then

$$\mathbb{E}[X] = (1+2+3+4+5) \cdot \frac{1}{10} + 6 \cdot \frac{1}{2} = \frac{15}{10} + 3 = 4.5.$$

The heavy mass on the face 6 has pulled the mean a full unit above the fair-die value of 3.5.

- (iv) *Sum of two dice (the laborious way).* For the sum $S = a + b$ of two fair dice on $\Omega = \{1, \dots, 6\}^2$, the value-by-value formula gives

$$\mathbb{E}[S] = \sum_{k=2}^{12} k \cdot \mathbb{P}(S = k) = 7,$$

after some bookkeeping over the eleven possible sums and their multiplicities. The next section will deliver a much cleaner two-line derivation of the same answer.

- (v) *First entry of a random permutation.* A **random permutation** of $\{1, \dots, n\}$ means picking one of the $n!$ orderings of these numbers uniformly at random; that is, $\Omega = S_n$

is the set of all orderings with the uniform measure. For an ordering $\sigma \in S_n$, we write $\sigma(i)$ for the entry that σ places at position i . Let $X = \sigma(1)$, the very first entry. For each $k \in \{1, \dots, n\}$, the orderings with $\sigma(1) = k$ are obtained by filling the remaining $n - 1$ positions with the other $n - 1$ values in any order, giving $(n - 1)!$ such orderings, so

$$\mathbb{P}(X = k) = \frac{(n - 1)!}{n!} = \frac{1}{n}.$$

Thus X is itself uniform on $\{1, \dots, n\}$, and

$$\mathbb{E}[X] = \sum_{k=1}^n k \cdot \frac{1}{n} = \frac{n+1}{2}.$$

Position 1 played no special role in this argument, so the same answer holds for $\sigma(2), \dots, \sigma(n)$: the entry at any single fixed position has expected value $(n + 1)/2$.

- (vi) *Fixed points of a random permutation of three letters.* Take $\Omega = S_3$, the six orderings of $\{1, 2, 3\}$, with the uniform measure. Call an index i a **fixed point** of σ if $\sigma(i) = i$ (the value i ends up in position i). Let $X(\sigma)$ be the number of fixed points of σ . Listing the six orderings as $\sigma(1)\sigma(2)\sigma(3)$ and counting fixed points in each:

$$123 \mapsto 3, \quad 132 \mapsto 1, \quad 213 \mapsto 1, \quad 231 \mapsto 0, \quad 312 \mapsto 0, \quad 321 \mapsto 1.$$

So X takes the value 0 twice, the value 1 three times, and the value 3 once, giving

$$\mathbb{E}[X] = 0 \cdot \frac{2}{6} + 1 \cdot \frac{3}{6} + 3 \cdot \frac{1}{6} = \frac{6}{6} = 1.$$

A random shuffle of three labels fixes one label in place on average. Strikingly, we will see in the next section that the same answer — exactly 1 fixed point on average — holds for shuffles of n labels for *every* n , and that the route through linearity is far less laborious than the enumeration here.

2.3 Averaging a function of a random variable (Optional.) The expectation is defined only for *real-valued* random variables. Yet many of the variables we care about are not numbers at all: a uniformly random vertex of a graph, or a uniformly random ordering of $\{1, \dots, n\}$, has a target set with no arithmetic on it, and asking for “the average vertex” is meaningless. What rescues us is that we rarely want the average of such an object directly; we want the average of some *number read off from it* — the degree of the random vertex, the number of fixed points of the random ordering. In each case there is a function f that turns the (possibly non-numerical) value into a real number, and it is $f(X)$ whose average we seek.

Concretely, suppose X is a random variable with target set S , and $f: S \rightarrow \mathbb{R}$ is any function. Then the composition $f(X)$, whose value on the outcome ω is $f(X(\omega))$, is a *real-valued* random variable, so its expectation is defined by the basic formula. The point of the next identity is that we never have to descend to the outcomes ω : the distribution of X alone — the masses $\mathbb{P}(X = s)$ — already determines $\mathbb{E}[f(X)]$.

Expectation of a function of a random variable. Let X be a random variable on a finite probability space $(\Omega, \mathbb{P})$ with target set S , and let $f: S \rightarrow \mathbb{R}$ be any function. Then $f(X)$ is a real-valued random variable, and

$$\mathbb{E}[f(X)] = \sum_{s \in S} f(s) \cdot \mathbb{P}(X = s).$$

Proof. Start from the defining formula for the expectation of the real-valued variable $f(X)$, which sums over outcomes:

$$\mathbb{E}[f(X)] = \sum_{\omega \in \Omega} f(X(\omega)) \cdot \mathbb{P}(\omega).$$

Now group the outcomes according to the value X reads on them. For each $s \in S$, the outcomes with $X(\omega) = s$ are exactly those forming the event $\{X = s\}$, and on every one of them the summand $f(X(\omega))$ equals the single number $f(s)$. Collecting each group into one term,

$$\mathbb{E}[f(X)] = \sum_{s \in S} \sum_{\omega : X(\omega) = s} f(s) \cdot \mathbb{P}(\omega) = \sum_{s \in S} f(s) \sum_{\omega : X(\omega) = s} \mathbb{P}(\omega) = \sum_{s \in S} f(s) \cdot \mathbb{P}(X = s),$$

since the inner sum of the masses of all outcomes in $\{X = s\}$ is precisely $\mathbb{P}(X = s)$. ■

Two observations make this identity easier to use. First, when $S \subseteq \mathbb{R}$ and f is the identity function $f(s) = s$, the formula collapses to the value-by-value formula $\mathbb{E}[X] = \sum_x x \mathbb{P}(X = x)$ from the start of the section — so the present identity is its natural generalisation, with an arbitrary f in place of “multiply by the value itself.” Second, nothing in the statement requires X itself to be real-valued. The variable X may take vertices, orderings, or any other values as long as f lands in $\mathbb{R}$; the average $\mathbb{E}[f(X)]$ is then computed entirely from the distribution of X , with the underlying Ω left out of sight. This is what lets us speak of, say, the average degree of a random vertex even though a vertex is not a number — though when the values of X are themselves the outcomes of the experiment, as for a uniformly random vertex, one can equally just read $\mathbb{E}[f(X)]$ off the definition directly, as we will do for the degree-sum formula below.

As a quick numerical check, let X be the number on a fair six-sided die and take $f(s) = s^2$. The identity gives

$$\mathbb{E}[X^2] = \sum_{s=1}^6 s^2 \cdot \frac{1}{6} = \frac{1 + 4 + 9 + 16 + 25 + 36}{6} = \frac{91}{6},$$

a number we will recognise on Day 5, when functions like X^2 return to measure how spread out a random variable is.

Exercise 2.2.1. A fair four-sided die shows one of the numbers 1, 2, 3, 4, each with probability 1/4. Let X be the number shown. Compute $\mathbb{E}[X]$.

Exercise 2.2.2. Let X be the number of heads in two fair coin tosses, with the distribution found in the previous section’s exercises. Use the value-by-value formula to compute $\mathbb{E}[X]$.

Exercise 2.2.3. A random variable X takes the value 2 with probability 1/3 and the value 5 with probability 2/3. Compute $\mathbb{E}[X]$, and confirm that X never actually takes this value.

2.3 INDICATOR RANDOM VARIABLES

2.4 The indicator of an event. Among all random variables, the simplest — and the one we will use most often — is the *indicator* of an event. Its job is to convert a yes/no question (“did the event A occur?”) into a 0-or-1 readout, so that the machinery of expectation can be brought to bear on it.

Definition. Let $A \subseteq \Omega$ be an event in a probability space $(\Omega, \mathbb{P})$. The **indicator** of A is the random variable $\mathbf{1}_A : \Omega \rightarrow \mathbb{R}$ defined by

$$\mathbf{1}_A(\omega) = \begin{cases} 1 & \text{if } \omega \in A, \\ 0 & \text{if } \omega \notin A. \end{cases}$$

Thus $\mathbf{1}_A$ records whether the event A occurred (1) or not (0).

The expectation of an indicator is the probability of the corresponding event — a small fact, but the engine of almost every calculation we will do this week.

Indicator-expectation identity. For any event A in a finite probability space,

$$\mathbb{E}[\mathbf{1}_A] = \mathbb{P}(A).$$

Proof. The variable $\mathbf{1}_A$ takes the value 1 on A and 0 on A^c , so the value-by-value expectation reads

$$\mathbb{E}[\mathbf{1}_A] = 1 \cdot \mathbb{P}(A) + 0 \cdot \mathbb{P}(A^c) = \mathbb{P}(A). \quad \blacksquare$$

Two quick checks lock in the identity.

- (i) *Heads of a fair coin.* On $\Omega = \{\mathbf{H}, \mathbf{T}\}$ with the uniform measure, let $A = \{\mathbf{H}\}$. Then $\mathbf{1}_A(\mathbf{H}) = 1$ and $\mathbf{1}_A(\mathbf{T}) = 0$, so

$$\mathbb{E}[\mathbf{1}_A] = 1 \cdot \frac{1}{2} + 0 \cdot \frac{1}{2} = \frac{1}{2} = \mathbb{P}(A).$$

- (ii) *Even roll of a die.* On $\Omega = \{1, \dots, 6\}$ with the uniform measure, let $A = \{2, 4, 6\}$. The indicator $\mathbf{1}_A$ reads 1 on three of the six outcomes, so $\mathbb{E}[\mathbf{1}_A] = 3 \cdot (1/6) = 1/2 = \mathbb{P}(A)$.

The identity is trivial to verify, but its real power emerges only when combined with the next ingredient.

Exercise 2.3.1. A fair die is rolled, and $A = \{3, 6\}$ is the event “the roll is a multiple of 3”. Write down the value of $\mathbf{1}_A(\omega)$ for each outcome ω , compute $\mathbb{E}[\mathbf{1}_A]$ directly, and check that it equals $\mathbb{P}(A)$.

Exercise 2.3.2. Let A be an event in a finite probability space. Show that $\mathbf{1}_{A^c}(\omega) = 1 - \mathbf{1}_A(\omega)$ holds for every outcome $\omega \in \Omega$.

Exercise 2.3.3. Let A and B be events in a finite probability space. Show that $\mathbf{1}_{A \cap B}(\omega) = \mathbf{1}_A(\omega) \mathbf{1}_B(\omega)$ for every outcome $\omega \in \Omega$.

2.4 LINEARITY OF EXPECTATION

2.5 The single most useful fact in the workshop. The expectation operator is *linear*, in a strong sense that no other operation in probability shares: it does not care whether the random variables involved interact in complicated ways.

Linearity of expectation. Let X and Y be real-valued random variables on the same finite probability space, and let $a, b \in \mathbb{R}$. Then

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y].$$

More generally, for any finite collection $X_1, \dots, X_n$ of real-valued random variables on the same probability space and any scalars $a_1, \dots, a_n \in \mathbb{R}$,

$$\mathbb{E}[a_1X_1 + a_2X_2 + \dots + a_nX_n] = a_1\mathbb{E}[X_1] + a_2\mathbb{E}[X_2] + \dots + a_n\mathbb{E}[X_n].$$

Setting all $a_i = 1$ recovers the additive form: the expectation of a sum is the sum of the expectations.

Proof. Unwind the definition and rearrange:

$$\begin{aligned} \mathbb{E}[aX + bY] &= \sum_{\omega} (aX(\omega) + bY(\omega))\mathbb{P}(\omega) \\ &= a \sum_{\omega} X(\omega)\mathbb{P}(\omega) + b \sum_{\omega} Y(\omega)\mathbb{P}(\omega) = a\mathbb{E}[X] + b\mathbb{E}[Y]. \end{aligned}$$

The general n -variable form follows by induction on n . ■

A first warm-up illustrates the power. We find the expected sum of the outcomes of two fair dice. Roll two fair dice, and let X_1, X_2 be the values shown. Each X_i has expectation 3.5 (the fair-die calculation), so by linearity,

$$\mathbb{E}[X_1 + X_2] = \mathbb{E}[X_1] + \mathbb{E}[X_2] = 3.5 + 3.5 = 7.$$

Compare this two-line derivation with the eleven-term sum in the previous section — same answer, far less labor. The hidden algebra of the eleven-term calculation is precisely what linearity buys us in one stroke.

2.6 A non-probabilistic question, solved probabilistically. One of the genuinely surprising features of linearity — and a small foreshadowing of everything we will do later in the week — is that it can be turned on questions that have nothing to do with probability on the surface. Here is a small instance.

What is the sum of all three-digit numbers whose digits lie in $\{1, 2, \dots, 9\}$, with repetition of digits allowed?

The question is entirely deterministic — there is a definite list of numbers, and a definite sum. No coin is being tossed, no permutation drawn, no graph randomized. The list, in lexicographic order, runs

$$111, 112, 113, \dots, 998, 999,$$

and there are $9^3 = 729$ entries (nine choices per position). Adding the 729 numbers by hand is unattractive; let us instead make a deliberate detour through probability.

Pick one of the 729 numbers *uniformly at random*, and write its digits, from left to right, as D_1, D_2, D_3 . The chosen number itself is

$$N = 100 D_1 + 10 D_2 + D_3,$$

which is exactly the kind of sum linearity was built to digest. For each position $i \in \{1, 2, 3\}$ and each value $v \in \{1, \dots, 9\}$, the number of entries on our list with $D_i = v$ is $9^2 = 81$ (the other two positions are free), so each D_i is uniform on $\{1, \dots, 9\}$ and

$$\mathbb{E}[D_i] = \frac{1 + 2 + \dots + 9}{9} = \frac{45}{9} = 5 \quad (i = 1, 2, 3).$$

By linearity,

$$\mathbb{E}[N] = 100 \mathbb{E}[D_1] + 10 \mathbb{E}[D_2] + \mathbb{E}[D_3] = 100 \cdot 5 + 10 \cdot 5 + 5 = 555.$$

Now for the bridge back to the original question. The expected value of a uniformly chosen item from a finite list is, by definition, the average of the list:

$$\mathbb{E}[N] = \frac{1}{729} \sum_{n \text{ on the list}} n,$$

so the sum we wanted is exactly

$$\sum_{n \text{ on the list}} n = 729 \cdot \mathbb{E}[N] = 729 \cdot 555 = 404\,595.$$

Two things are worth noticing about this argument.

- *The sum was not in the problem; we put it there.* The original question gave us a list of numbers and asked for their total. We picked one of those numbers at random, decomposed it as $N = 100 D_1 + 10 D_2 + D_3$, and applied linearity to that decomposition. The place-value identity, which has nothing to do with probability, is what made the random variable N available as a sum of three simpler ones.
- *Probability acted as a bookkeeping device.* Behind the scenes, the calculation $\mathbb{E}[N] = 100 \cdot 5 + 10 \cdot 5 + 5$ is a reorganized way of saying “each digit 1, 2, ..., 9 appears in each of the three positions exactly 81 times.” One could in principle reach the answer 404 595 by that direct tally. What the random-variable language buys us is a single uniform vocabulary — expectation, linearity, marginal distribution — in which the bookkeeping happens almost automatically, with no per-problem cleverness required.

Throughout the week, we will see this template recur on increasingly serious questions: a list (of graphs, of subsets, of colorings, of sequences) we wish to understand is replaced by a single uniformly chosen element of it, properties of that element are reframed as random variables, and linearity is unleashed. In every case, the only probability in sight is the one we introduce.

2.7 The indicator decomposition. The most common shape of a “linearity” argument at this workshop is the following. If X records how many of the events $A_1, \dots, A_n$ occur, then we can write X as a sum of indicators,

$$X = \mathbf{1}_{A_1} + \mathbf{1}_{A_2} + \dots + \mathbf{1}_{A_n},$$

and so by linearity together with the indicator-expectation identity,

$$\mathbb{E}[X] = \sum_{i=1}^n \mathbb{E}[\mathbf{1}_{A_i}] = \sum_{i=1}^n \mathbb{P}(A_i).$$

Read this twice. To compute the expected number of events from a given list that occur, we only need to know the individual probabilities — we do not need to understand how the events interact. This is the workshop’s main manoeuvre, and it will run, with minor variations, for the rest of the week.

- (i) *Number of heads in n tosses.* Toss n fair coins independently. Let A_i be the event “the i -th toss is heads”; then the number of heads is $X = \mathbf{1}_{A_1} + \dots + \mathbf{1}_{A_n}$. Each $\mathbb{P}(A_i) = 1/2$, so

$$\mathbb{E}[X] = \sum_{i=1}^n \mathbb{P}(A_i) = \sum_{i=1}^n \frac{1}{2} = \frac{n}{2}.$$

No surprise — but notice that the calculation never invoked the binomial distribution; the answer fell out of a one-line application of linearity.

- (ii) *Fixed points of a random permutation.* Let σ be a uniformly random permutation of $\{1, 2, \dots, n\}$. A **fixed point** of σ is an index i with $\sigma(i) = i$. Let $A_i = \{\sigma(i) = i\}$; the number of fixed points is then $X = \mathbf{1}_{A_1} + \dots + \mathbf{1}_{A_n}$. By symmetry, each $\mathbb{P}(A_i) = (n-1)!/n! = 1/n$, so

$$\mathbb{E}[X] = \sum_{i=1}^n \mathbb{P}(A_i) = n \cdot \frac{1}{n} = 1.$$

On average, a random permutation of n objects has exactly one fixed point, no matter how large n is. The events A_i are *not* independent (this was the content of Problem 8 of Set 1) — but the linearity calculation does not care.

- (iii) *Adjacent friend-pairs around a circular table.* Seat n people uniformly at random around a circular table, and suppose m of the $\binom{n}{2}$ pairs of them are friends. What is the expected number of friend-pairs that end up sitting next to each other? For each friend-pair $\{u, v\}$, let $A_{\{u,v\}}$ be the event “ u and v are seated next to each other”. By the seating-symmetry argument (fix u ’s seat without loss of generality; v then lands uniformly in one of the remaining $n-1$ seats, exactly 2 of which are adjacent to u), $\mathbb{P}(A_{\{u,v\}}) = 2/(n-1)$. The number of adjacent friend-pairs is $\sum_{\{u,v\}} \mathbf{1}_{A_{\{u,v\}}}$, so by linearity,

$$\sum_{\{u,v\}} \mathbb{P}(A_{\{u,v\}}) = m \cdot \frac{2}{n-1} = \frac{2m}{n-1}.$$

The events $A_{\{u,v\}}$ are wildly correlated — moving a single person changes the adjacency status of every pair involving that person — but linearity, once again, does not care.

Exercise 2.4.1. Three fair dice are rolled. Using linearity of expectation, find the expected value of the sum of the three numbers shown.

Exercise 2.4.2. A fair coin is tossed ten times. Writing the number of heads as a sum of indicator random variables, find its expected value.

Exercise 2.4.3. Four fair dice are rolled. Find the expected number of dice that show a six. (Hint: write this count as a sum of four indicators, one per die.)

2.5 A FIRST APPLICATION: THE DEGREE-SUM FORMULA

2.8 Graphs and degrees. Before stating the identity, we recall the basic vocabulary of graphs.

Definition. A **graph** $G = (V, E)$ consists of a finite **vertex set** V and an **edge set** $E \subseteq \binom{V}{2}$, where $\binom{V}{2}$ denotes the set of all unordered pairs of distinct vertices. The **degree** of a vertex $v \in V$, written $\deg(v)$, is the number of edges in E containing v .

A quick sanity check before the general claim. In the triangle on three vertices, every vertex has degree 2, and $\sum_v \deg(v) = 2 + 2 + 2 = 6$, which is twice the number of edges (3). In a star with one centre joined to four leaves, the centre has degree 4 and each leaf has degree 1, so $\sum_v \deg(v) = 4 + 4 \cdot 1 = 8$, twice the number of edges (4). In both cases the sum of the degrees turned out to be twice the number of edges — a pattern we now prove in general, by computing one expectation in two ways.

Theorem 2.9. Degree-sum formula. Let $G = (V, E)$ be a finite graph. Then

$$\sum_{v \in V} \deg(v) = 2|E|.$$

Proof. Turn the vertex set into a probability space: take $\Omega = V$ with the uniform measure $\mathbb{P}(\{v\}) = 1/|V|$ for every $v \in V$, so the experiment is simply “pick a vertex of G at random.” On this space the degree function $\deg: V \rightarrow \mathbb{R}$ is itself a real-valued random variable — each outcome is a vertex, and $\deg$ reads off its degree. We compute its expectation $\mathbb{E}[\deg]$ in two ways, and equate the answers.

The first way is to read the expectation straight off the definition as a sum over outcomes. Since the outcomes are the vertices and $\mathbb{P}(\{v\}) = 1/|V|$,

$$\mathbb{E}[\deg] = \sum_{v \in V} \deg(v) \cdot \mathbb{P}(\{v\}) = \frac{1}{|V|} \sum_{v \in V} \deg(v). \quad (*)$$

The second way decomposes the degree into indicators. For each edge $e \in E$, let A_e be the event that the chosen vertex is one of the two endpoints of e . Because the degree of a vertex is exactly the number of edges containing it, counting those edges one at a time gives

$$\deg = \sum_{e \in E} \mathbf{1}_{A_e},$$

where the indicator $\mathbf{1}_{A_e}$ reads 1 on each of the two endpoints of e and 0 on every other vertex. Each edge contains exactly two of the $|V|$ vertices, so $\mathbb{P}(A_e) = 2/|V|$. By linearity of expectation and the indicator-expectation identity $\mathbb{E}[\mathbf{1}_{A_e}] = \mathbb{P}(A_e)$,

$$\mathbb{E}[\deg] = \sum_{e \in E} \mathbb{E}[\mathbf{1}_{A_e}] = \sum_{e \in E} \mathbb{P}(A_e) = \sum_{e \in E} \frac{2}{|V|} = \frac{2|E|}{|V|}. \quad (**)$$

Equating $(*)$ and $(**)$ and multiplying through by $|V|$ gives the claim. $\blacksquare$

Remark. The identity has an equally short combinatorial proof: count the pairs (v, e) with $v \in e$ in two ways. The two proofs are essentially the same; the probabilistic version simply makes “two ways of counting” formal, by computing one expectation in two ways. This is a pattern we will see again — it is the same idea Sperner’s theorem will rest on tomorrow.

Exercise 2.5.1. Verify the degree-sum formula directly for the 4-cycle (a square: four vertices, each joined to its two neighbours). Compute $\sum_v \deg(v)$ and $2|E|$ separately, and check that they agree.

Exercise 2.5.2. A graph has 10 vertices, every one of degree 3. Use the degree-sum formula to determine its number of edges.

Exercise 2.5.3. Using the degree-sum formula, show that in any finite graph the number of vertices of odd degree is even.

2.6 THE KEY MOVE: COMPUTING EXPECTATIONS

2.10 A heuristic to internalise. Whenever the task is to compute $\mathbb{E}[X]$ for a random variable X that counts some quantity, the standard manoeuvre is:

- (1) Write X as a sum of indicators, $X = \sum_i \mathbf{1}_{A_i}$, for an explicit list of events A_i .
- (2) Compute each $\mathbb{P}(A_i)$ separately.
- (3) Apply linearity: $\mathbb{E}[X] = \sum_i \mathbb{P}(A_i)$.

The manoeuvre is so reliable that it is worth memorizing as a reflex. A slightly more elaborate worked example will illustrate the pattern end-to-end.

Example 2.11 (Coupon collector, partial form). There are n types of coupons. Each day, one coupon arrives, of a type chosen uniformly at random from the n types and independently of all previous days. After m days, what is the expected number of *distinct* types collected?

Let X be the number of distinct types after m days, and for each type $i \in \{1, \dots, n\}$, let A_i be the event “type i has appeared at least once in the m days”. The number of distinct types collected is exactly the number of these events that occurred, so

$$X = \mathbf{1}_{A_1} + \mathbf{1}_{A_2} + \dots + \mathbf{1}_{A_n}.$$

The complement of A_i is the event “type i never appears”, which has probability $(1 - 1/n)^m$ (since the m choices are independent, each avoiding type i with probability $1 - 1/n$). So $\mathbb{P}(A_i) = 1 - (1 - 1/n)^m$, and by linearity,

$$\mathbb{E}[X] = \sum_{i=1}^n \mathbb{P}(A_i) = n \cdot \left(1 - \left(1 - \frac{1}{n}\right)^m\right).$$

Two sanity checks: $m = 0$ gives $\mathbb{E}[X] = 0$ (nothing has arrived yet), and as $m \rightarrow \infty$ the inner $(1 - 1/n)^m$ tends to 0, so $\mathbb{E}[X] \rightarrow n$ (eventually everything is collected). The events A_i share a common arrival stream and so are emphatically not independent — but the linearity step never asks whether they are.

Example 2.12 (Same-suit neighbours in a shuffled deck). A standard deck of 52 cards is shuffled uniformly at random. What is the expected number of positions $i \in \{1, \dots, 51\}$ at which the cards in positions i and $i + 1$ belong to the same suit?

Let A_i be the event “positions i and $i + 1$ hold cards of the same suit”, and let X be the count we want, so that

$$X = \mathbf{1}_{A_1} + \mathbf{1}_{A_2} + \dots + \mathbf{1}_{A_{51}}.$$

For $\mathbb{P}(A_i)$ we use a symmetry argument: fix attention on whichever card lands in position i , and ask what the card in position $i + 1$ is. Among the 51 remaining cards, exactly 12 share the suit of the position- i card, and the position- $i + 1$ card is equally likely to be any of those 51. So

$$\mathbb{P}(A_i) = \frac{12}{51},$$

for every $i \in \{1, \dots, 51\}$. By linearity,

$$\mathbb{E}[X] = \sum_{i=1}^{51} \mathbb{P}(A_i) = 51 \cdot \frac{12}{51} = 12.$$

On average, a shuffled deck has twelve same-suit adjacent pairs. The events A_i are wildly correlated — consecutive positions share a card, so A_i and A_{i+1} talk about a common card — but the calculation is unbothered. Once again the manoeuvre runs on marginal probabilities only.

Exercise 2.6.1. A fair die is rolled 12 times. Find the expected number of rolls that show a six.

Exercise 2.6.2. In a group of n people, each person's birthday is a uniformly random day of a 365-day year, independently of the others. Find the expected number of people whose birthday falls on 1 January.

Exercise 2.6.3. A standard deck of 52 cards is shuffled uniformly at random. Find the expected number of aces among the top five cards.

2.7 INDEPENDENCE OF RANDOM VARIABLES (OPTIONAL)

2.13 From events to random variables. On Day 1 we said that two *events* A and B are independent when $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$. The same idea — one quantity carries no information about the other — has a counterpart for random variables, and it is one of the most important structural conditions on a pair (or tuple) of random variables.

An intuitive picture first. Two coin tosses performed in physically separate corners of the room are independent: the second toss does not know what the first one did. The two rolls of a pair of fair dice are independent: the spots showing on one die put no constraint on the spots showing on the other. By contrast, the entries $\sigma(1)$ and $\sigma(2)$ of a uniformly random permutation are *not* independent — knowing $\sigma(1) = 3$ forces $\sigma(2) \neq 3$, which shifts the distribution of $\sigma(2)$. And the most extreme failure of independence is the diagonal: X and X itself. Knowing X tells you everything about X .

The clean formal definition turns “carries no information” into a factorisation of joint probabilities into marginals.

Definition. Independent random variables Let X and Y be random variables defined on the same finite probability space. We say X and Y are **independent** if for every pair of values x, y in their respective target sets,

$$\mathbb{P}(X = x \text{ and } Y = y) = \mathbb{P}(X = x) \cdot \mathbb{P}(Y = y).$$

More generally, the random variables $X_1, X_2, \dots, X_n$ on a common probability space are **mutually independent** if for every choice of values $x_1, x_2, \dots, x_n$,

$$\mathbb{P}(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n) = \prod_{i=1}^n \mathbb{P}(X_i = x_i).$$

Two ways of reading this definition that will both be useful.

In the language of events. The random variables $X_1, \dots, X_n$ are mutually independent precisely when, for *every* choice of values $x_1, \dots, x_n$, the events $\{X_1 = x_1\}, \dots, \{X_n = x_n\}$ are mutually independent in the Day 1 sense. So independence of random variables is genuinely no more than “independence of every value-event tuple at once”.

In the language of distributions. Independence says that the joint distribution of $(X_1, \dots, X_n)$ is completely pinned down by the n individual distributions: any joint probability is recovered by multiplying the relevant marginals. Most joint distributions look nothing like this — but the experiments we describe procedurally as “independent draws” (independent coin tosses,

independent dice rolls, independently chosen edges in $G(n, p)$) are designed to produce exactly this structure.

2.14 Examples and non-examples. Three illustrations to fix the notion.

- (i) *Two independent fair coin tosses.* Take $\Omega = \{H, T\}^2$ with the uniform measure, and let X_i be the outcome of the i -th toss. Each of the four outcomes has probability $1/4$. So, for example, $\mathbb{P}(X_1 = H, X_2 = H) = 1/4 = (1/2)(1/2)$, and the same factorisation holds for the other three pairs of values. The two tosses are independent.
- (ii) *Two entries of a random permutation.* Take $\Omega = S_n$ with the uniform measure, $n \geq 2$, and let $X_i = \sigma(i)$. Each X_i is uniform on $\{1, \dots, n\}$, so $\mathbb{P}(X_i = k) = 1/n$ for every k . But $\mathbb{P}(X_1 = k, X_2 = k) = 0$ (no permutation places k at two different positions), whereas $\mathbb{P}(X_1 = k)\mathbb{P}(X_2 = k) = 1/n^2 \neq 0$. The factorisation fails, and X_1 and X_2 are *not* independent.
- (iii) *A variable and itself.* For any non-constant X , the pair (X, X) is not independent. Joint probabilities live on the “diagonal” — $\mathbb{P}(X = x, X = y) = 0$ unless $x = y$ — while the product of marginals $\mathbb{P}(X = x)\mathbb{P}(X = y)$ is generally non-zero off the diagonal. Knowing one copy tells us everything about the other.

2.15 The product rule for independent variables. Linearity told us that the expectation of a sum is the sum of the expectations, no matter how the summands interact. For *products* no unconditional rule holds, but independence is exactly the condition that buys us the analogous statement.

Product rule for independent variables. If X and Y are independent real-valued random variables on the same probability space, then

$$\mathbb{E}[XY] = \mathbb{E}[X] \cdot \mathbb{E}[Y].$$

More generally, if $X_1, X_2, \dots, X_n$ are mutually independent, then

$$\mathbb{E}[X_1 X_2 \cdots X_n] = \mathbb{E}[X_1] \mathbb{E}[X_2] \cdots \mathbb{E}[X_n].$$

Proof. Apply the value-by-value formula to XY and use independence to factor the joint probability:

$$\mathbb{E}[XY] = \sum_{x,y} xy \cdot \mathbb{P}(X = x, Y = y) = \sum_{x,y} xy \cdot \mathbb{P}(X = x) \mathbb{P}(Y = y).$$

The double sum factors as a product of single sums,

$$\sum_{x,y} xy \cdot \mathbb{P}(X = x) \mathbb{P}(Y = y) = \left(\sum_x x \cdot \mathbb{P}(X = x) \right) \left(\sum_y y \cdot \mathbb{P}(Y = y) \right) = \mathbb{E}[X] \cdot \mathbb{E}[Y].$$

The n -variable case follows by induction. ■

The product rule *requires* independence: without it, $\mathbb{E}[XY]$ can be very different from $\mathbb{E}[X]\mathbb{E}[Y]$. The simplest failure is again the diagonal one. If $Y = X$ and X is non-constant, then $\mathbb{E}[XY] = \mathbb{E}[X^2]$ while $\mathbb{E}[X]\mathbb{E}[Y] = (\mathbb{E}[X])^2$, and these two differ by exactly $\mathbb{E}[X^2] - (\mathbb{E}[X])^2$ — a quantity that will play the leading role on Day 5 (it is the *variance* of X).

2.16 A note on terminology: “i.i.d.” One phrase is so common it deserves naming up front. A finite sequence $X_1, X_2, \dots, X_n$ of random variables is called **independent and identically distributed** — almost always abbreviated **i.i.d.** — when (a) the X_i are mutually independent

and (b) they all share a common distribution. The canonical example is “toss n fair coins independently”: each X_i is uniform on $\{H, T\}$, and the tosses are independent. Whenever we speak of “independent copies” of a random variable in the days ahead, we mean i.i.d. copies.

Exercise 2.7.1. Two fair dice are rolled independently, showing values X and Y . Verify the factorisation $\mathbb{P}(X = 2, Y = 5) = \mathbb{P}(X = 2)\mathbb{P}(Y = 5)$.

Exercise 2.7.2. For the same two independent dice, use the product rule to compute $\mathbb{E}[XY]$.

Exercise 2.7.3. Let X be the value shown by a single fair die, and set $Y = X$. Compute $\mathbb{E}[XY] = \mathbb{E}[X^2]$ and $\mathbb{E}[X]\mathbb{E}[Y] = (\mathbb{E}[X])^2$ separately, and observe that they differ — so the product rule can fail when the two variables are not independent.

PROBLEM SET 2

Today’s problem set is mostly about practising the indicator-and-linearity move on a variety of structures. After this set, the manoeuvre should feel automatic.

Problem 2.7.1. A fair coin is tossed 10 times. Let X be the number of heads. Compute $\mathbb{E}[X]$ in two ways:

- (a) By the value-by-value formula $\mathbb{E}[X] = \sum_k k \mathbb{P}(X = k)$, using $\mathbb{P}(X = k) = \binom{10}{k}/2^{10}$.
- (b) By writing X as a sum of indicators and using linearity.

Problem 2.7.2. n letters are placed at random into n pre-addressed envelopes (one letter per envelope). What is the expected number of letters that end up in their correct envelope?

Problem 2.7.3. n balls are dropped independently and uniformly at random into m boxes. Let X be the number of *empty* boxes after all balls have been placed. Show that

$$\mathbb{E}[X] = m \left(1 - \frac{1}{m}\right)^n.$$

Problem 2.7.4. Let $G = (V, E)$ be a finite graph and let S be a uniformly random subset of V (each vertex is included independently with probability $1/2$). Compute the expected number of edges $\{u, v\} \in E$ for which both u and v lie in S .

Problem 2.7.5. A sequence of n symbols, each chosen uniformly and independently from $\{0, 1, 2, \dots, 9\}$, is generated. Compute the expected number of digits in $\{0, 1, \dots, 9\}$ that appear at least once in the sequence.

Problem 2.7.6. Let σ be a uniformly random permutation of $\{1, 2, \dots, n\}$. An index i is called an *ascent* if $\sigma(i) < \sigma(i+1)$. Compute the expected number of ascents.

Problem 2.7.7 * Suppose n couples are seated uniformly at random in a row of $2n$ chairs. Compute the expected number of couples who end up sitting next to each other.

Problem 2.7.8 * Let $G = (V, E)$ be a finite graph in which every vertex has the same degree d (such graphs are called *d-regular*). Use the degree-sum formula to express $|E|$ in terms of $|V|$ and d . Then deduce that $d \cdot |V|$ must be even.

Problem 2.7.9. A uniformly random permutation σ of $\{1, 2, \dots, n\}$ is chosen. Call position i a *record* (or *left-to-right maximum*) if $\sigma(i)$ is larger than every earlier value: that is, $\sigma(i) > \sigma(j)$ for all $j < i$. (Position 1 is always a record; position 2 is a record exactly when $\sigma(2) > \sigma(1)$; and so on.)

Compute the expected number of records.

Hint: For a fixed i , focus on the relative order of just the first i values $\sigma(1), \sigma(2), \dots, \sigma(i)$. Among these, where is the largest equally likely to sit?

Problem 2.7.10. A fair coin is tossed n times, producing a string $s_1 s_2 \dots s_n \in \{H, T\}^n$. A *run* of s is a maximal block of consecutive tosses of the same outcome. For example, the string HHTHTTT has four runs: HH, T, H, TTT.

Compute the expected number of runs in a uniformly random length- n coin-flip string.

Problem 2.7.11 * n guests at a party (with $n \geq 3$) are seated around a circular table. Each guest brought a wrapped gift, which they tossed into a pile. After dinner, the pile is shuffled uniformly at random and one gift is returned to each guest (so the assignment of gifts to guests is a uniformly random permutation).

Call a guest *pleased* if they receive either their own gift or the gift brought by one of their two table-neighbours. Compute the expected number of pleased guests.

APPENDIX A

A PRIMER ON GRAPHS

Several of the workshop’s most striking results — the bipartite cut bound of Day 3, the Ramsey lower bounds of Days 3 and 4, Turán’s theorem on Day 4 — speak about *graphs*: about vertices, edges, cliques, independent sets, and colourings. This appendix collects the modest amount of graph-theoretic vocabulary the workshop uses, builds it from the ground up, and illustrates each piece with a small drawing.

The diagrams follow the visual conventions of Douglas West’s *Introduction to Graph Theory*: a vertex is a small solid disc, an edge is a thin line segment between two such discs, and a vertex-name sits just outside its dot.

A.1 GRAPHS, VERTICES, AND EDGES

A.1 The basic object. A graph records relationships among a fixed collection of objects: which pairs are tied together by some chosen relation, and which are not. Drawn on paper, the objects become dots and the ties become lines; stripped of the picture, the underlying structure is just two sets — a set of dots and a set of dot-pairs.

Definition. A **graph** G is a pair (V, E) in which V is a finite non-empty set called the set of **vertices**, and E is a set of two-element subsets of V , called **edges**. If $\{u, v\} \in E$ we say that u and v are **adjacent** (or **neighbours**); the edge $\{u, v\}$ is then said to be **incident** to each of u and v .

The number of vertices $|V|$ is the **order** of G , and the number of edges $|E|$ is the **size** of G . We will routinely write $V(G)$ and $E(G)$ when several graphs are in play at once.

A few features of the definition deserve emphasis before we draw anything. First, an edge is an unordered pair: $\{u, v\}$ is the same edge as $\{v, u\}$. There is no preferred direction along an edge — in this workshop, all graphs are *undirected*.

Second, edges are subsets of size exactly two. In particular, $\{u, u\}$ is not a legal edge (no loops), and no two distinct edges of G share the same pair of endpoints (no parallel edges). Graphs satisfying these two restrictions are sometimes called *simple* graphs; in this workshop “graph” always means “simple graph”.

Third, V and E are abstract sets: the picture is a convenience, not part of the data. Two drawings of the same (V, E) that differ only in how the dots are arranged on the page represent the same graph. Figure A.1 shows one such drawing.

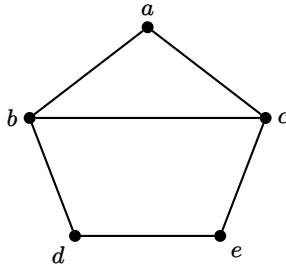


Figure A.1: A graph G with vertex set $V = \{a, b, c, d, e\}$ and edge set $E = \{\{a, b\}, \{a, c\}, \{b, c\}, \{b, d\}, \{c, e\}, \{d, e\}\}$. Vertices a and b are adjacent; vertices a and d are not. The edge $\{a, b\}$ is incident to both a and b .

The same graph could be drawn with a at the bottom and d, e at the top, or with the edges curved instead of straight: the diagram would look different, but (V, E) would be identical, and so would every theorem we proved about it.

A.2 Notation in the workshop. A handful of conventions, used throughout these notes:

- We write $n = |V|$ for the number of vertices and $m = |E|$ for the number of edges when no confusion arises.
- An edge $\{u, v\}$ is sometimes abbreviated to uv , with the understanding that uv and vu name the same edge.
- The notation $u \sim v$ is shorthand for “ u and v are adjacent”, i.e. $\{u, v\} \in E$.

None of this is deep. It is the standard vocabulary one picks up after a single lecture; we list it here only so that nothing in the main text comes as a surprise.

A.2 DEGREE AND THE HANDSHAKE LEMMA

A.3 Counting an edge at each endpoint. The most useful single number attached to a vertex is the count of how many edges touch it.

Definition. Degree The **degree** of a vertex v in a graph G is the number of edges of G that are incident to v , or equivalently the number of vertices of G adjacent to v . It is written $\deg_G(v)$, or simply $\deg(v)$ when the underlying graph is clear from context. The **average degree** of G is

$$\bar{d} = \frac{1}{n} \sum_{v \in V} \deg(v)$$

where $n = |V|$.

For the graph of Figure A.1, a direct count gives $\deg(a) = 2$, $\deg(b) = 3$, $\deg(c) = 3$, $\deg(d) = 2$, $\deg(e) = 2$. Figure A.2 labels each vertex with its degree. That last numerical coincidence in the caption — the degrees adding to twice the edge count — is not a coincidence at all. It is the first non-trivial fact about graphs, and the proof is a clean count-twice argument.

A.4 The handshake lemma. The following result was published by Euler in 1736 in the context of the Königsberg bridges problem, often regarded as the founding paper of graph theory.

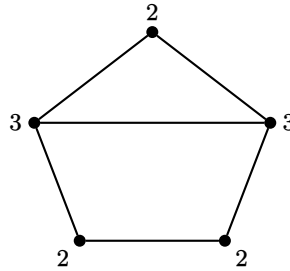


Figure A.2: The graph of Figure A.1 with each vertex labelled by its degree. The degrees sum to $2 + 3 + 3 + 2 + 2 = 12$, which is exactly twice the number of edges.

Handshake lemma. For any graph $G = (V, E)$,

$$\sum_{v \in V} \deg(v) = 2|E|.$$

Equivalently, the average degree is $\bar{d} = 2|E|/|V|$.

Proof. Count the pairs (v, e) with $v \in V$, $e \in E$, and v incident to e , in two ways. On one hand, fixing v and ranging over e , the number of incident pairs is $\deg(v)$, so the total is $\sum_v \deg(v)$.

On the other hand, fixing e and ranging over v , every edge $e = \{u, w\}$ contributes exactly two incidences (one at u , one at w), so the total is $2|E|$. Equating the two expressions for the same count gives the lemma. ■

The picturesque name comes from a re-reading: at a party of n guests, let V be the guests and let $\{u, v\}$ be an edge if u and v shook hands. Then $\deg(v)$ is the number of handshakes v took part in, and the lemma says: the total number of handshakes recorded by all guests equals twice the number of distinct handshakes that actually occurred — because every handshake is recorded once by each participant. A first corollary: in any graph, the number of vertices of odd degree is even.

A.3 SOME STANDARD GRAPHS

The constructions of Days 3 and 4 (random colourings of complete graphs, random subgraphs, random orderings of vertices) rely on a small zoo of named graphs that recur throughout. Each is most easily understood by sight.

A.5 The complete graph K_n . A vertex set with *every* pair joined by an edge defines the densest possible graph.

Definition. Complete graph For an integer $n \geq 1$, the **complete graph** K_n has n vertices, and its edge set consists of *all* $\binom{n}{2}$ two-element subsets of V .

In K_n every vertex is adjacent to every other vertex, so each vertex has degree $n - 1$. The handshake lemma then confirms $|E(K_n)| = n(n - 1)/2 = \binom{n}{2}$, which the definition also gives directly. Figure A.3 shows K_3 , K_4 , K_5 in the standard drawings (vertices arranged on a regular polygon, all $\binom{n}{2}$ edges as chords). The two-colourings of K_n in Day 3's Ramsey argument are precisely 2-colourings of these $\binom{n}{2}$ edges; the graph itself never changes.

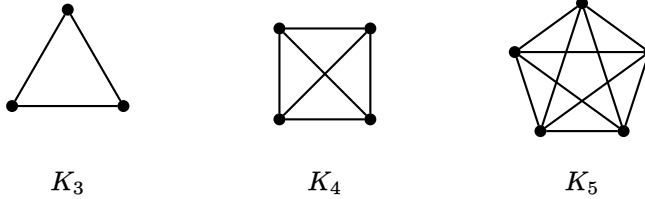


Figure A.3: The first few complete graphs. K_3 is the triangle; K_4 adds the two diagonals to a square; K_5 has $\binom{5}{2} = 10$ edges and is famously *non-planar* (every drawing in the plane has at least one edge-crossing).

A.6 Paths and cycles. Two other named families show up so often that they deserve standing names.

Definition. Let $n \geq 1$. A graph $G = (V, E)$ is a **path** P_n if $|V| = n$ and one can list its vertices in a sequence $v_1, v_2, \dots, v_n$ (each vertex of G appearing exactly once) so that the edges are *precisely* the consecutive pairs and nothing else:

$$E = \{v_i v_{i+1} : 1 \leq i \leq n-1\}.$$

Let $n \geq 3$. A graph $G = (V, E)$ is a **cycle** C_n if $|V| = n$ and one can list its vertices in a sequence $v_1, v_2, \dots, v_n$ (each appearing exactly once) so that the edges are precisely the consecutive pairs together with the single pair $v_n v_1$ closing the loop, and nothing else:

$$E = \{v_i v_{i+1} : 1 \leq i \leq n-1\} \cup \{v_n v_1\}.$$

The phrasing “one can list its vertices” deserves a second look, because it is what makes these definitions clean. The labels $v_1, \dots, v_n$ are not part of the graph; they are a bookkeeping device we are free to choose. To say a graph *is* a path is to assert that *some* such ordering of its vertices exists — a structural property of the graph, not a feature of how its vertices happen to be named. The same graph can usually be listed in several valid ways (for a path, reading the sequence backwards is one), and any one of them witnesses that it is a path. The clause “and nothing else” is doing real work too: it forbids any stray edge, so the graph contains exactly the $n-1$ edges of the chain (or the n edges of the loop) and no more. Pictures, with $n=5$, in Figure A.4.

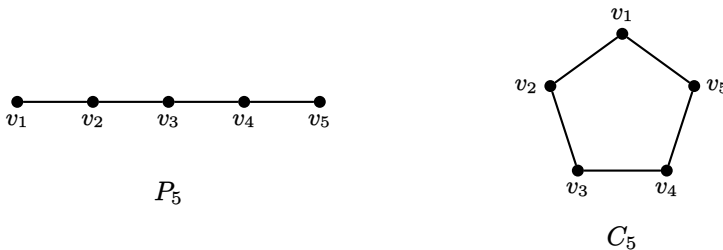


Figure A.4: The path P_5 and the cycle C_5 . The path has two vertices of degree 1 (the endpoints) and $n-2$ vertices of degree 2; in a cycle, every vertex has degree exactly 2.

The triangle $C_3 = K_3$ is the smallest cycle. Day 4’s problems on triangle-free graphs and 4-cycle-free graphs are statements about which finite graphs do, or do not, contain C_3 or C_4 as a subgraph.

A.7 Bipartite and complete bipartite graphs. A graph is **bipartite** if its vertices can be split into two camps so that every edge has one endpoint in each camp.

Definition. A graph $G = (V, E)$ is **bipartite** if there exists a partition $V = A \sqcup B$ such that every edge of G has one endpoint in A and the other in B . The pair (A, B) is called a **bipartition** of G .

For positive integers m, n , the **complete bipartite graph** $K_{m,n}$ has vertex set $A \sqcup B$ with $|A| = m$ and $|B| = n$, and its edges are *all* mn pairs $\{a, b\}$ with $a \in A$ and $b \in B$.

Figure A.5 shows a generic bipartite graph and the complete bipartite graph $K_{3,3}$.

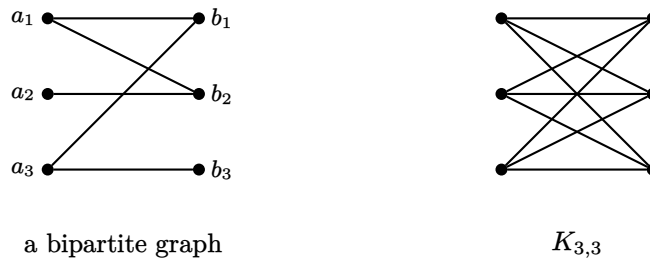


Figure A.5: Left: a bipartite graph on parts $A = \{a_1, a_2, a_3\}$ and $B = \{b_1, b_2, b_3\}$; the five edges all cross from A to B . Right: the complete bipartite graph $K_{3,3}$, in which *every* pair (a_i, b_j) is an edge.

Day 3's bipartite-cut theorem is a statement about turning an arbitrary graph into a bipartite spanning subgraph: given any G , one can choose a partition $V = A \sqcup B$ so that at least half the edges cross from A to B . The complete bipartite graphs $K_{m,n}$ themselves are the densest bipartite graphs and appear in several extremal problems — for instance, $K_{2,2} = C_4$, the four-cycle, is the obstacle that “ C_4 -free” graphs must avoid.

A small but useful test: K_n is bipartite only when $n \leq 2$; for $n \geq 3$, K_n contains a triangle, and a triangle cannot be bipartite (any 2-colouring of three pairwise-adjacent vertices forces two same-coloured ones to be adjacent). More generally, a graph is bipartite if and only if it contains no *odd* cycle as a subgraph.

A.4 SUBGRAPHS AND COMPLEMENTS

A.8 Subgraphs. A subgraph is what you get by deleting some vertices and/or some edges. Two natural variants come up repeatedly in the main text.

Definition. A graph $H = (V', E')$ is a **subgraph** of $G = (V, E)$, written $H \subseteq G$, if $V' \subseteq V$ and $E' \subseteq E$ (and, as required for E' to be a legal edge set, every edge of E' has both endpoints in V').

The subgraph H is **induced** if $E' = \{\{u, v\} \in E : u, v \in V'\}$ — that is, E' contains *every* edge of G whose endpoints both lie in V' . In that case we write $H = G[V']$ and call it the **subgraph of G induced by V'** .

Two subgraphs of the same vertex set $V' \subseteq V$ can differ: an arbitrary subgraph is free to omit edges that the induced subgraph keeps. Figure A.6 compares the two for a small example.

Two language conventions worth noting. A subgraph H is **spanning** if $V(H) = V(G)$ — it uses every vertex, but is free to drop edges. The bipartite-cut theorem of Day 3 produces a

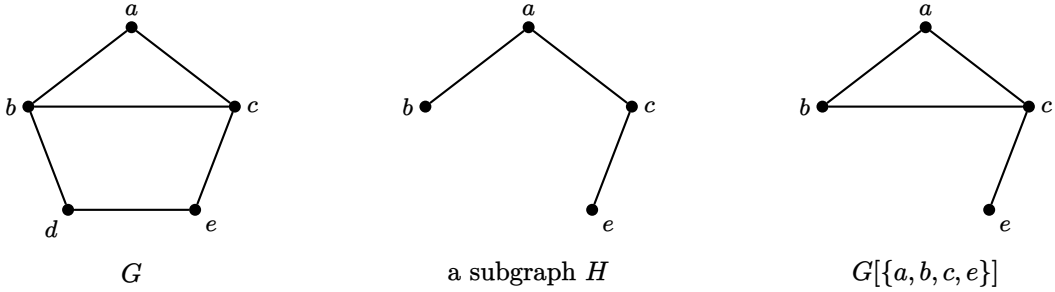


Figure A.6: Left: the original graph G . Middle: a subgraph H on vertex set $\{a, b, c, e\}$ that omits the edge $\{b, c\}$ — legal, since subgraphs may drop edges. Right: the *induced* subgraph $G[\{a, b, c, e\}]$ on the same vertex set; the edge $\{b, c\}$ must be kept because both its endpoints survive.

spanning bipartite subgraph of G . And the term “subgraph” alone, without qualifiers, always means “not necessarily induced, not necessarily spanning”.

A.9 The complement. A graph and its complement encode the same vertex set with opposite ideas about which pairs are tied.

Definition. The **complement** of a graph $G = (V, E)$ is the graph $\bar{G} = (V, \bar{E})$ on the same vertex set, with edge set

$$\bar{E} = \{\{u, v\} : u, v \in V, u \neq v, \{u, v\} \notin E\}.$$

In other words, u and v are adjacent in $\bar{G}$ if and only if they are *not* adjacent in G .

Equivalently, G and $\bar{G}$ together carry every edge of K_n : their edge sets partition $E(K_n)$. Figure A.7 illustrates this on a five-vertex example.

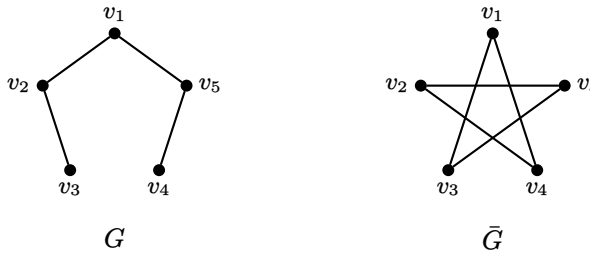


Figure A.7: A graph G and its complement $\bar{G}$ on the same five vertices. Every pair of distinct vertices is an edge in exactly one of the two graphs; together they realise all $\binom{5}{2} = 10$ edges of K_5 . Here G has 4 edges, so $\bar{G}$ has $10 - 4 = 6$.

The complement is the workhorse of the swap “large independent set in $G \Leftrightarrow$ large clique in $\bar{G}$ ” that we used to deduce Turán’s original theorem from the independent-set version on Day 4. We turn to those two notions next.

A.5 CLIQUES AND INDEPENDENT SETS

A.10 The two opposite extremes. A set of vertices can be “maximally tied together” or “maximally untied”. Both are useful.

Definition. Let $G = (V, E)$ be a graph. A subset $S \subseteq V$ is a **clique** of G if every two distinct vertices in S are adjacent in G , i.e. $G[S] = K_{|S|}$.

A subset $I \subseteq V$ is an **independent set** of G if no two vertices in I are adjacent in G , i.e. $G[I]$ has no edges.

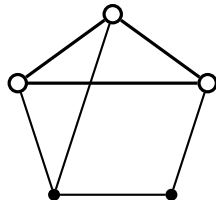
The **clique number** $\omega(G)$ is the size of the largest clique in G ; the **independence number** $\alpha(G)$ is the size of the largest independent set in G .

A clique in G is exactly an independent set in $\bar{G}$ and vice versa, since S being a clique in G means “every pair within S is an edge of G ”, which means “every pair within S is a non-edge of $\bar{G}$ ”. So

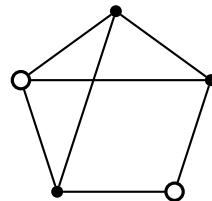
$$\omega(G) = \alpha(\bar{G}), \quad \alpha(G) = \omega(\bar{G}).$$

This is the duality that powers the complement-swap in Turán’s theorem.

Figure A.8 marks a 3-clique and a 3-independent set inside a single graph; the same diagram can be read either way, since the marked vertices are simply chosen subsets.



a clique $\{a, b, e\}$



an independent set $\{b, d\}$

Figure A.8: The same graph on five vertices, with two different vertex subsets highlighted (white discs). On the left, the three highlighted vertices form a clique — all three edges between them are present in G (shown thicker). On the right, the two highlighted vertices form an independent set: no edge of G joins them. Here $\omega(G) = 3$ and $\alpha(G) = 2$.

The whole point of Turán’s theorem on Day 4 is to give a lower bound on $\alpha(G)$ that depends only on the average degree of G — and, by the complementary reading, an upper bound on the number of edges a graph can have without containing K_{r+1} .

A.6 EDGE COLOURINGS

A.11 Painting the edges. Day 3’s Ramsey argument is phrased as a statement about colouring the edges of K_n red or blue. The set-up is general enough to deserve a definition.

Definition. Let $G = (V, E)$ be a graph and let C be a finite set, whose elements we call **colours**. An **edge colouring** of G with colour set C is any function $\chi: E \rightarrow C$.

A subset $S \subseteq V$ is **monochromatic** (under χ) if every two distinct vertices of S are adjacent in G and all $\binom{|S|}{2}$ edges of $G[S]$ receive the same colour.

No restriction on χ is required: in particular, adjacent edges may share a colour. The version of edge colouring that demands distinct colours on every two edges sharing a vertex — “proper edge colouring” — is a different topic and not used in this workshop.

Figure A.9 shows a two-colouring of K_4 with red and blue (drawn solid and dashed, respectively).

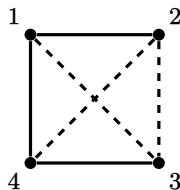


Figure A.9: A 2-colouring of the six edges of K_4 : solid edges are red, dashed edges are blue. The triangle $\{1, 2, 4\}$ has its three edges coloured red, red, red and is therefore monochromatic; the triangle $\{1, 2, 3\}$ has edges coloured red, blue, blue and is not.

The Ramsey lower bounds of Days 3 and 4 are existence statements about edge 2-colourings of K_n : “there exists a colouring with no monochromatic K_k ”. The randomness sits on the colouring (each edge picks its colour by a coin flip), and the count is of how many k -subsets $S \subseteq V$ end up monochromatic.

A.7 GLOSSARY AT A GLANCE

A.12 Vocabulary recap. The terms below are the ones the workshop’s main text uses without further explanation. Each was defined in an earlier section of this appendix; the table is a one-page reference card.

- Working vocabulary.**
- (i) **Graph** $G = (V, E)$. A finite vertex set V and a set E of two-element subsets of V (edges). No loops, no parallel edges.
 - (ii) **Adjacency**. $u \sim v$ means $\{u, v\} \in E$.
 - (iii) **Degree**. $\deg(v)$ = number of edges incident to v . Average degree $\bar{d} = 2|E|/|V|$.
 - (iv) **Handshake lemma**. $\sum_v \deg(v) = 2|E|$.
 - (v) **Subgraph** $H \subseteq G$. A graph obtained from G by possibly removing vertices and/or edges. **Induced** subgraphs keep every G -edge whose endpoints survive.
 - (vi) **Complement** $\bar{G}$. Same vertex set as G , with edges exactly the non-edges of G .
 - (vii) **Clique**. A vertex set with every pair adjacent; the largest size is $\omega(G)$.
 - (viii) **Independent set**. A vertex set with no two vertices adjacent; the largest size is $\alpha(G)$.
 - (ix) **Bipartite graph**. Vertex set splits as $V = A \sqcup B$ with every edge crossing.
 - (x) **Complete graph** K_n . Every pair of vertices is an edge; $\binom{n}{2}$ edges in total.
 - (xi) **Complete bipartite graph** $K_{m,n}$. Parts of sizes m, n with every cross-pair an edge; mn edges in total.
 - (xii) **Path** P_n , **cycle** C_n . The straight-line graph and the polygonal graph on n vertices, respectively.
 - (xiii) **Edge colouring**. A function $\chi: E \rightarrow C$. A vertex set S is **monochromatic** if every edge of $G[S]$ receives the same χ -value.

This is the entire vocabulary the workshop needs. Whenever a Day's text says "Let G be a graph with n vertices and average degree d , suppose G has no triangle, ...", every clause unpacks into the definitions above. Readers seeing graph theory for the first time may find it useful to redraw Figures A.1–A.9 by hand on scratch paper; the muscle memory of placing dots and joining them is what makes the rest of these notes look like pictures rather than symbols.